Mid. Farah Awad

Theory of Computation

Topic 1

* Mathmatical Preliminaries :-
① Sets ② Function ③ Relation ④ Graphs ⑤ Proof of techniques

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* Sets is a Collection of elements . مجموعة من العناصر

ex: A = {1, 2, 3} — set

B = {bus, train, cars} — set

bus ∈ B ✓

- does 4 ∈ A ? No 4 ∉ A

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* Set operations :

① Membership : ∈ Belongs to تنتمي ; ∉ not belongs to لا تنتمي	② Union . اتحاد ∪ 	③ Intersection تقاطع ∩ 
④ Difference الفرق − , \ or	⑤ Complement متمم $\bar{S}$ or S'	⑥ Subset ⊆ proper ⊂
⑦ Power Set 2^S , $P(S)$	⑧ Set Cardinality (size) |S| عدد العناصر	⑨ Cartesian Product A X B

* Difference between List and Set :

List : ordering ; may Duplicate

Set : No Duplicate

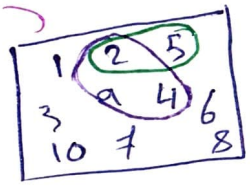
* Set Representations

① Finite set	② Infinite set
$S = \{a, b, c, \dots, k\}$ II $S = \{a, b, c, d, e, f, g, h, k\}$	$S = \{2, 4, 6, \dots\}$ $S = \{j \mid j \geq 0 \text{ and } j = 2k \text{ and } k \geq 0\}$ such that

* Universal sets

All possible elements . U

ex: U



$U = \{1, 2, 3, \dots, 10\}$

$A = \{2, 4, 9\}$

$B = \{2, 5\}$

① $A \cup B = \{2, 4, 5, 9\}$ (اتحاد، تجمع)

② $A \cap B = \{2\}$

③ $A \cap U = A$

④ $A \cup U = U$

⑤ $A - B = \{4, 9\}$

⑥ $B - A = \{5\}$

⑦ $\bar{A} = \{1, 3, 5, 6, 7, 8, 10\}$

⑧ $\bar{U} = \emptyset$

⑨ $\bar{B} = \{1, 3, 4, 6, 7, 8, 9, 10\}$

⑩ $|\Phi| = 0$

⑪ $|A| = 3$

⑫ $|B| = 2$

⑬ $|A \cap B| = 1$

⑭ $|\bar{B} \cap A| = \{4, 9\} = 2$

⑮ $P(\{1\}) = \{\Phi, \{1\}\}$

⑯ $P(\{0, 1\}) = \{\Phi, \{0\}, \{1\}, \{0, 1\}\}$

example ① of Power set

given $S = \{1, 2, 3\}$

$2^S = P(S) = \{\Phi, \{1\}, \{2\}, \{3\}, \{1, 2\}, \{1, 3\}, \{2, 3\}, \{1, 2, 3\}\}$

$|2^S| = P(S) = 2^{|S|} = 2^3 = 8$

ple ② Power set:

given $|A| = 4$

- Find $|2^A| = 2^{|A|} = 2^4 = \boxed{16}$

- Find $|P(A)| = 2^{|A|} = 2^4 = \boxed{16}$

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example ③

Find $|P(P\{a\})|$

let $x = P\{a\} = \{\Phi, \{a\}\}$

then $P(x) = \{\Phi, \{\Phi\}, \{\Phi, a\}, \{a\}\}$.

$P(P\{a\}) = 2^{|2^{\{a\}}|} = 2^{|2^{\{a\}}|} = 2^2 = \boxed{4}$

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* Cartesian Product:

ex ① given $A = \{1\}$, $B = \{2\}$

$A \times B = \{(1,2)\}$

one element

ex ② given $X = \{a, b\}$, $Y = \{c, d, e\}$

① $X \times Y = \{(a,c), (a,d), (a,e), (b,c), (b,d), (b,e)\}$

② $|X \times Y| = |X| * |Y| = 2 * 3 = 6$

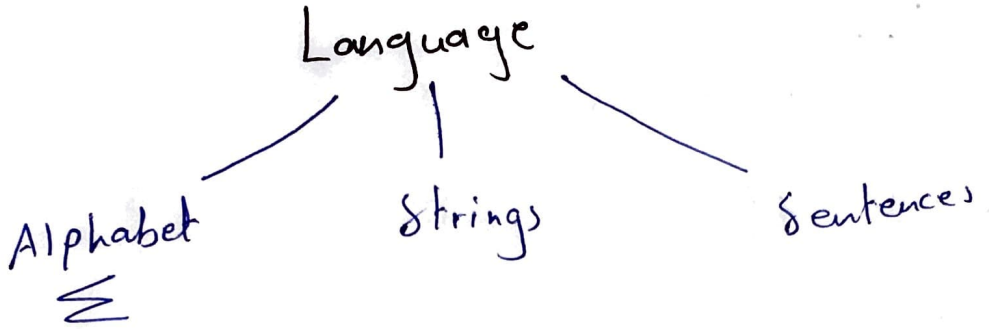
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* disjoint set

$A \cap B = \emptyset$

الـ A و B ما فيها عناصر مشتركة

* Language:
is a set of words (strings):
└ a sequence of symbols from some Alphabet.
Σ



* example:

strings : cat, dog, home

Language : {cat, dog, home}

Alphapet = Σ : {a, b, c, ..., z}

Languages is used to describe computation problems.
is translated to set membership

* Alphabet: is a set of symbols:

Σ = {0, 1} Binary.

Σ = {0, 1, 2, ..., 9}

Σ = {A, B, C, ..., Z, a, b, c, ..., z}

Σ_{unary} = {1}

Σ_{octal} = {0, 1, 2, ..., 7}

Language unary
ال لغة

* String Operations :-

① Concatenation :

$w = a_1 a_2 \dots a_n$ $\quad$ $u = b_1 b_2 \dots b_m$

$w.u = wu = a_1 a_2 \dots a_n b_1 b_2 \dots b_m$

String length

$|w| = n$

عدد العناصر اللي جوه

ex Find $w = 11$, $u = \{1\}$ $\quad$ $\Sigma = \{1\}$

$wu = 111$

ex $\Sigma \{a, b\}$, $w = aba$ $\quad$ $u = bba$

Find

① $uw = bbaaba$

② $ww = ababa$

③ $uuw = bba\,bba\,aba$

④ $|uw| = |u| + |w| = 3+3 = 6$ (size)

⑤ $wu = ababba$

⑥ $|uuw| = |u| + |u| + |w| = 9$

② String Reverse

$w = a_1 a_2 \dots a_n$ $\quad$ abcabbb ←

$w^R = a_n, a_{n-1} \dots a_1$ $\quad$ bbbaba

نعكسهم

③ Empty String

A string with no letters denoted. λ or ϵ

- $|\lambda| = 0$
- $\lambda w = w\lambda = w$

يعني موجودين ما بتأثر وجودها وعدمها واحد.

- $\lambda abba = abba\lambda = a\lambda bba = abba$

④ substring

- substring of string, a subsequence of consecutive characters.

String	Substrig
abbab	ab
	abb
	abba
	bbab
	b

Does baa substring of abbab
No, لازم يكونو ورى بعض

⑤ Prefix and suffix

$w = uv$

prefix suffix

ex: ① prefix of abbab and suffix. (word: abbab)

prefix = λ , suffix : abbab.

ex ② prefix and sufix of bbab (word: abbab)

prefix = a , suffix = bbab

⑥ Exponent operation

w^n = wwww..w (n times)

w: string, n: integer

$w^0 = \lambda$

ex: $(abba)^2 = abbaabba$.

$(bba)^0 = \lambda$

⑦ The * operation the kleene closure

The set of all possible string from Σ

$\Sigma^* = \Sigma^0 \cup \Sigma^1 \cup \Sigma^2 \cup \dots$

$\Sigma^* = \bigcup_{i=0}^{\infty} \Sigma^i$

Example ① The * operation :

assume $\Sigma = \{1\}$

Find Σ^* ?

$$\Sigma^* = \bigcup_{i=0}^{\infty} \Sigma^i = \Sigma^0 \cup \Sigma^1 \cup \Sigma^2 \cdots$$

$$= \{\lambda\} \cup \{1\} \cup \{11\} \cup \{111\} \cdots$$

$$= \{\lambda, 1, 11, 111, 1111, \ldots\}$$

لأنه احنا بنعاطلوع string

Example ②: assume $\Sigma = \{a, b\}$

Find Σ^* ?

$$\Sigma^* = \{\lambda\} \cup \{a, b\} \cup \{ab, aa, bb, ba\} \cup$$

$$\{aaa, aab, aba, abb, baa, bab, bba, bbb\}$$

$$= \{\lambda, a, b, aa, ab, ba, bb, aaa, aab, \ldots\}$$

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③ The + operation the postive closure.

- the set of all possible strings from Σ except λ.

λ الا Σ من strings كل

Example: assume $\Sigma = \{a, b\}$

Find Σ^+ ?

$$\Sigma^+ = \Sigma^* - \lambda$$

$$\Sigma^+ = \{a, b, aa, ab, \ldots\}$$

قصدهاي بس بدون λ

* Languages :-

A Language over alphabet Σ is any subset of Σ^*

Σ^* هي subset من

example :

(a) assume $\Sigma = \{a\}$

does these examples of Languages ?

① $L = \{\}$ ✓ Language

② $L = \{\lambda\}$ ✓ Language

③ $L = \{a, aa\}$ ✓ Language

④ $L = \{\lambda, a\}$ ✓ Language

⑤ $L = \{a\}$ ✓ Language

⑥ $L = \{a, aa, aaa\}$ ✓ Language

⑦ $L = \{a, b\}$ ✗ Not Language

⑧ $L = \{aa, bb\}$ ✗ Not Language

لازم يكون مضاف
فيه b , Σ^*
موجودة بالسؤال

(b) assume $\Sigma = \{a, b\}$

does these examples of Languages ?

① $L = \{ab, bba, bbab\}$ ✓ Language

② $L = \{a, ba\}$ ✓ Language

③ $L = \{a^n b^n : n \geq 1\}$ ✓ Language

$= \{ab, aabb, aaabbb, \dots\}$

$n=1$ حالة

$n=2$ حالة

ملاحظة :-
الأرقام ما بتعامل معهم حسب
الـ value بتعامل معهم كـ
string لأنه بتعامل مع character

* Prime Numbers :

$\Sigma = \{0, 1, 2, \dots, 9\}$

$L_{prime} = \{x : x \text{ is prime}\}$

$L_{prime} = \{2, 3, 5, 7, 11, 13, 17, \dots\}$

* Even Numbers

$L_{even} = \{0, 2, 4, 6, 8, 10, \dots\}$

$L_{even} = \{x : x \in \Sigma^* \text{ and } x \text{ is even}\}$

$\Sigma_{Addition} = \{1, +, =\}$

$L_{Addition} = \{ \underbrace{x+y=z}_{string} : x = 1^n, y = 1^m, z = 1^k, n+m=k, n, m, k \geq 1 \}$

example ①

ⓐ $n=1, m=1, k=2$

$1 + 1 = 11 \quad \in L_{Addition}$ Length = 6

ⓑ $n=3, m=2, k=5$

Length = 12

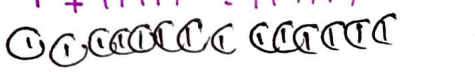
$111 + 11 = 11111$



ⓒ $n=1, m=5, k=6$

Length = 14

$1 + 11111 = 111111$



$\Sigma_{square} = \{1, \#\}$

$L_{square} = \{ x \# y : x = 1^n, y = 1^m, m = n^2 \}$

example ①

① $n=0$

$1^0 \# 1^0 = \lambda \# \lambda = \#$

② $n=1$

$1^1 \# 1^1 = 1 \# 1$

③ $n=2$

$1^2 \# 1^4 = 11 \# 1111$

④ $n=3$

$1^3 \# 1^9 = 111 \# 111111111$

"لغتنا" بتعامل مع الأرقام على أساس إنها string ما أخربط، أحطهم على أساس value

* Two Special Languages:

① Empty Language •

$|\{\}| = 0$ size = 0

② Language with Empty String

$|\{\lambda\}| =$ size = 1

Note: ① $\emptyset = \{\} \neq \{\lambda\}$

② set size $|\{\}| = |\emptyset| = 0$

③ set size $|\{\lambda\}| = 1$

④ String Length $|\lambda| = 0$

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* operations on Languages:

① $\{ab, ba\} \cup \{ab, bb\} = \{ab, ba, bb\}$

② $\{ab, ba\} \cap \{ab, bb\} = \{ab\}$

③ $\{ab, ba\} - \{ab, bb\} = \{ba\}$

* $L = \{ab, ba\}$

④ $\bar{L} = \Sigma^* - L = \{\lambda, a, b, aa, bb, \ldots\}$ كل الكلمات الي ماعدا L

⑤ $L^R = \{w^R : w \in L\}$

ex: ① $\{aab, bab, aaba\}^R = \{baa, bab, abaa\}$

② $\{a^n b^n : n \geq 1\}^R = \{b^n a^n : n \geq 1\}$

⑥ Concatenation: $L_1 L_2 = \{xy : x \in L_1, y \in L_2\}$

ex ①: $\{a, ba, bba\}\{ab, bb\}$

$= \{aab, abb, baab, babb, bbaab, bbabb\}$

② $\{a^n b^n : n \geq 1\}\{b\}$

$= \{a^n b^{n+1} : n \geq 1\}$

③ $\{a^n b : n \geq 0\}\{b^n a : n \geq 1\}$

$= \{a^k b b^m : k \geq 0, m \geq 1\}$

Exponent operation: L^n — positive integer, $L^n = LL \dots L$ (Language), by Def $L^0 = \{\lambda\}$

ex1) $\{ab, bb\}^2 = \{ab, bb\}\{ab, bb\}$
$= \{abab, abbb, bbab, bbbb\}$

ex2) $\{ab, bb\}^3 = \{ab, bb\}\{ab, bb\}\{ab, bb\}$
$= \{abab, abbb, bbab, bbbb\}\{ab, bb\}$
$= \{ababab, ababbb, abbbab, abbbbb, bbabab, bbabbb, abbbbab, bbbbbb\}$

ex3) assume $L = \{a^n b^n : n \geq 1\}$

① $L^0 = \{\lambda\}$

② $L^2 = \{a^n b^n : n \geq 1\}\{a^n b^n : n \geq 1\}$
$= \{a^m b^m a^k b^k : m, k \geq 1\}$

③ $L_3 = \{a^i b^i a^j b^j a^k b^k : i, j, k \geq 1\}$

variable: متغير

8) Star Closure (Kleene*):

$* = 0, 1, 2, 3, \dots \infty$

$L^* = \bigcup_{i=0}^{\infty} L^i$

ex) $\{a, bb\}^*$ L_0 L_1 L_2
$= \{\lambda, a, bb, aa, abb, bba, bbbb, aaa, aabb, abba, abbbb, bbaa, bbabb, bbbba, bbbbbb, \dots\}$

$\{a, bb\}^2 = \{a, bb\}\{a, bb\} = \{aa, abb, bba, bbbb\}$

9) positive Closure (the + operation)

$L^+ = \bigcup^{\infty} L^i$

$L^+ = L^* - \{\lambda\}$

ex) $\{a, bb\}^+ = \{a, bb, aa, \cancel{\quad}, abb, bba, bbbb, \dots\}$

$\{\lambda\}$ كلهم ما عدا

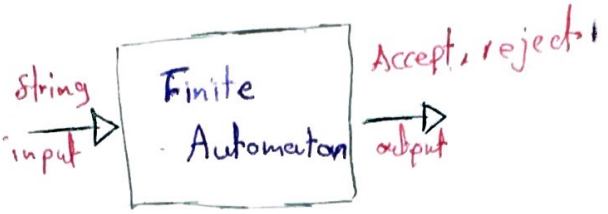
Mid

Theory of Computation

Topic 2

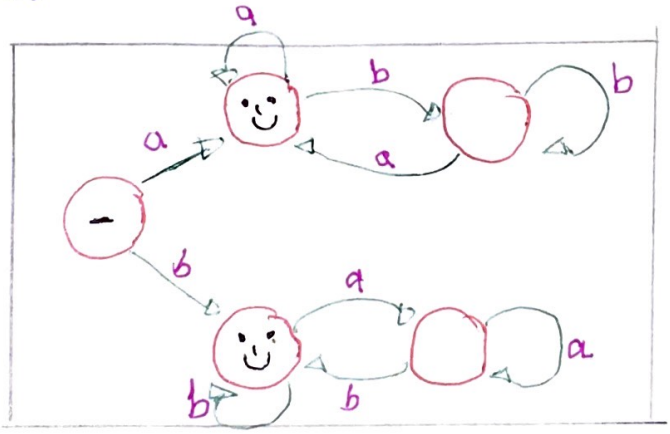
* Finite Automata
  - DFA → Deterministic Finite Automaton
  - NFA

* DFA



Language ... Graph

* Example (1)



start state

Final state

+ or ◎

or

* start state: ... نبدأ منها

* Final state: No أو yes بتشوف منها وهل الكلمة تتقبل إذا
اذا انتهت منها بتقبل yes إذا ما انتهت
منها بترفض No .

	Input	output
①	b a	No , reject
②	a	yes , accept
③	b	yes , accept
④	λ	No, reject
⑤	aab	No, reject
⑥	aaba	accept
⑦	bb a	reject
⑧	bbab	accept
⑨	aaa	accept
⑩	bbb	accept
⑪	abbba	accept
⑫	baaab	accept
⑬	aaaab	Reject
⑭	bbbb a	Reject

④ اذا كان عندي Start F مفروض يرجع accept فاضيا معناها No

الحل بناءً على الرسمة إلي قبل

① بس بمشي b بعدين a

بعدين ○ هاي معناها reject



انتهى الكون معناها reject

③ وبس بمشي a



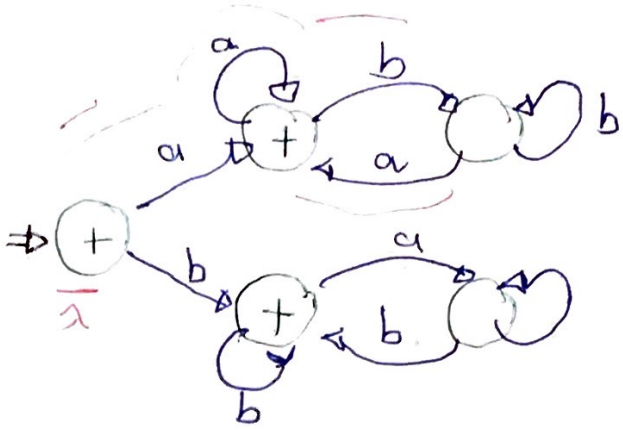
انتهى الكون معناها accept

وهكذا .

• دايماً نبلش من عند الـ start .

Example (2)

input	out put
aba	accept
aabb	reject
aabba	accept
λ	accept



(+) Final state

⇒(+) start State

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* Formal Definition of DFA:

$$M = (Q, \Sigma, S, q_0, F)$$

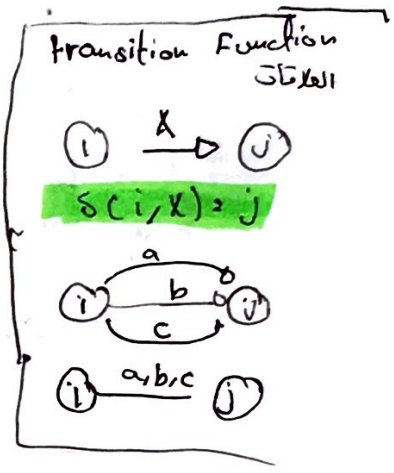
Q = Set of states

Σ: input of alphabet a, b, c - - -

S: Transition Function

q_0: initial state (start) →○ or (-) $q_0 \in Q$

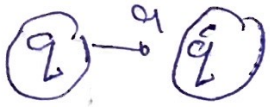
F: Final state ◎ $F \subseteq Q$



* Transition Function

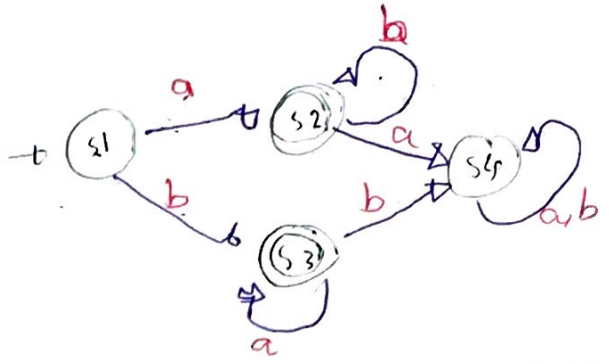
$$S(q, a) = \acute{q}$$

$q \in Q$, $a \in \Sigma$, $\acute{q} \in Q$



Example(1):

input	output
a	accept
b	accept
baa	accept
abcu	reject
bba	reject
λ	reject



- نلاحظ انه a وعدد n بأي من b accept
- نلاحظ انه b وعدد n بأي من a accept

$L(M) = \{ab^n, n \geq 0\} \cup \{ba^n = n \geq 0\}$

① $Q = \{s1, s2, s3, s4\}$

② $\Sigma = \{a, b\}$

③ δ .

① $\delta(s_1, a) = s2$
② $\delta(s_1, b) = s3$
③ $\delta(s_2, a) = s4$
④ $\delta(s_2, b) = s2$
⑤ $\delta(s_3, a) = s3$
⑥ $\delta(s_3, b) = s4$
⑦ $\delta(s_4, a) = s4$
⑧ $\delta(s_4, b) = s4$

δ	Σ
Q	Q

b $\delta : Q \times \Sigma \rightarrow Q$

	δ	a	b
start	s_1	s_2	s_3
final	s_2	s_4	s_2
final	s_3	s_3	s_4
	s_4	s_4	s_4

④ $q_0 = s_1$

⑤ $F = \{s_2, s_3\}$

لذا اعتبرنا انه الرسمة مش موجودة
فبدنا نطلع إذا reject, accept

	input	output
①	abba	reject
②	baa	accept
③	abbb	accept

abba حالة M حتى بنزل على s1 وبشوف علاقتها مع a بتعطينا s2 هيدين بنزل على s2 بشوف علاقتها مع b بتعطينا s4 وهيدين بنزل على s4 علاقتها مع b s4 بنزل s4 وبشوف علاقتها مع b

Example (2)

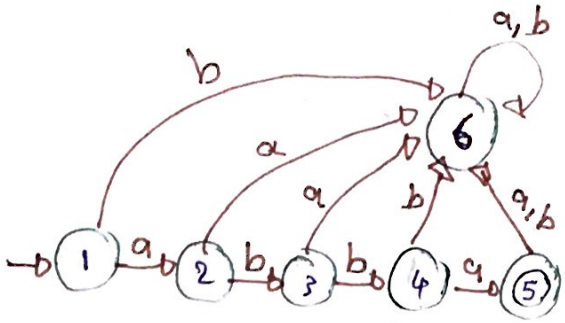
① $Q = \{1, 2, 3, 4, 5, 6\}$

② $\Sigma = a, b$

③ δ
① $\delta(1,a) = 2$ ② $(2,a) = 6$
② $\delta(1,b) = 6$ ③ $(2,b) = 3$
③ $\delta(3,a) = 6$ ④ $(3,b) = 4$
④ $\delta(4,a) = 5$ ⑤ $(4,b) = 6$
⑤ $\delta(5,a) = 6$ ⑥ $(5,b) = 6$
⑥ $\delta(6,a) = 6$ ⑦ $(6,b) = 6$
⑦ $\delta(1,abbbaa) = 6$

④ $Q_0 = 1$

⑤ $F = 5$



special case of δ

$\delta(q, \lambda) = q$

state may be repeated

	input	output
الدخل إلى accept يقبل	abba	accept
	aba	reject
	λ	reject

$L(M) = \{abba\}$ الـ strings اللي بتقبل accept

Example (2)

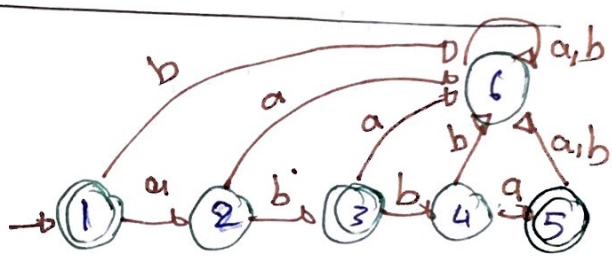
① $Q = \{1, 2, 3, 4, 5, 6\}$

② $\Sigma = \{a, b\}$

③ δ نفس المثال اللي فوق

④ $q_0 = 1$

⑤ $F = \{1, 3, 5\}$



input	output
λ	accept
ab	accept
abba	accept
abb	reject
ba	reject

$L(M) = \{\lambda, ab, abba\}$

Example (4)

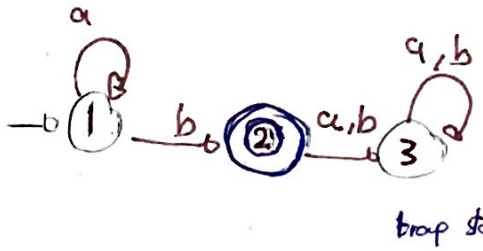
① Q = {1, 2, 3}

② Σ = {a, b}

③ δ = ① δ(1, a) = 1 ② δ(1, b) = 2
③ δ(2, a) = 3 ④ δ(2, b) = 3
⑤ δ(3, a) = 3 ⑥ δ(3, b) = 3

④ $q_0 = 1$

⑤ F = 2



input	output
b	accept
aab	accept
bab	reject
ab	accept

$L(M) = \{ a^n b, n \geq 0 \}$

Example (5)

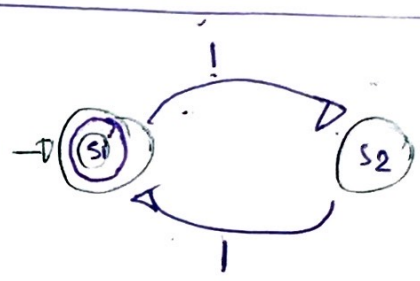
① Q = { δ_1, δ_2 }

② Σ = {1}

③ δ = ① δ(δ_1, 1) = δ_2 ② δ(δ_2, 1) = δ_1

④ $q_0 = \delta_1$

⑤ F = δ_1



input	output
1	reject
1 1	accept
1 1 1	reject
1 1 1 1	accept

Even = { x : x ∈ Σ^* and x even }

= { λ, 11, 1111, 1111 ... }

. 1. S all strings in $\{a,b\}^*$ containing at least

in general:

for DFA: $(Q, \Sigma, \delta, q_0, F)$

* Language accepted by M: (machine)

$$L(M) = \{ w \in \Sigma^* ; \delta^*(q_0, w) \in F \}$$



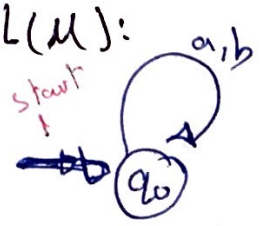
* Language rejected by M:

$$L(M) = \{ w \in \Sigma^* ; \delta^*(q_0, w) \notin F \}$$

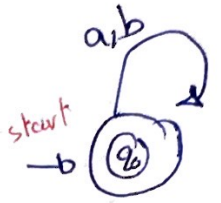


---

* Find L(M):

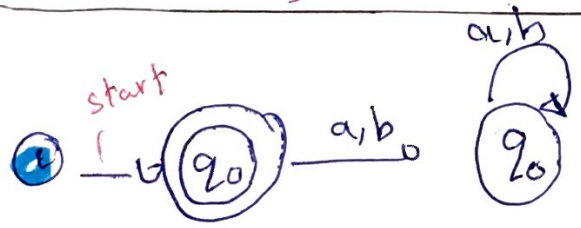
1. 

$L(M) = \{\}$ Empty

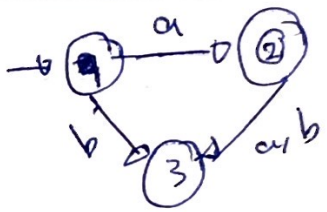
2. 

$L(M) = \{\lambda, a, ab, ba, aa \dots\}$

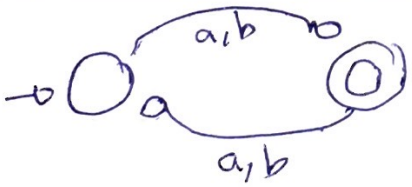
$L(M) = \Sigma^*$

3. 

$L(M) = \{\lambda\}$

4. 

$L(M) = \{a\}$

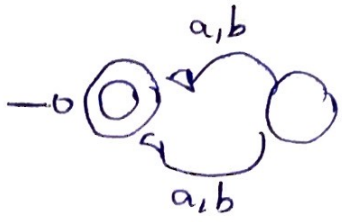
5. 

$L(M) = \{a, b, aba, aaa, \dots\}$

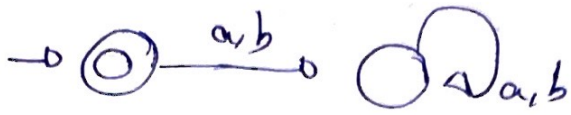
$L(M) = \{w \in \Sigma^* , |w| \bmod 2 = 1\}$

* Design DFA for $L = \{w : w \in \Sigma^*, |w| \bmod 2 = 0\}$

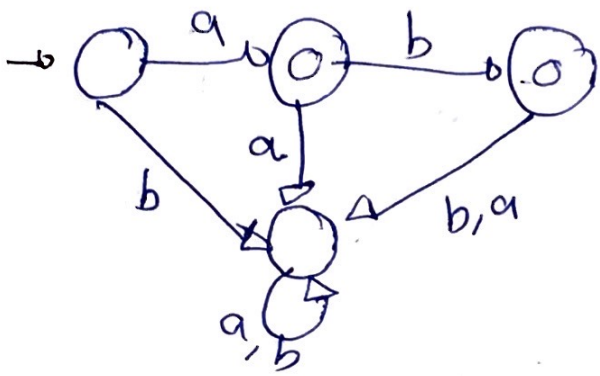
$\Sigma = \{a, b\}$ $\quad \Sigma^* = \{aa, bb, ab, ba, aaaa, bbbb \dots\}$



* Design DFA for $L = \{\lambda\}$ $\quad \Sigma = \{a, b\}$

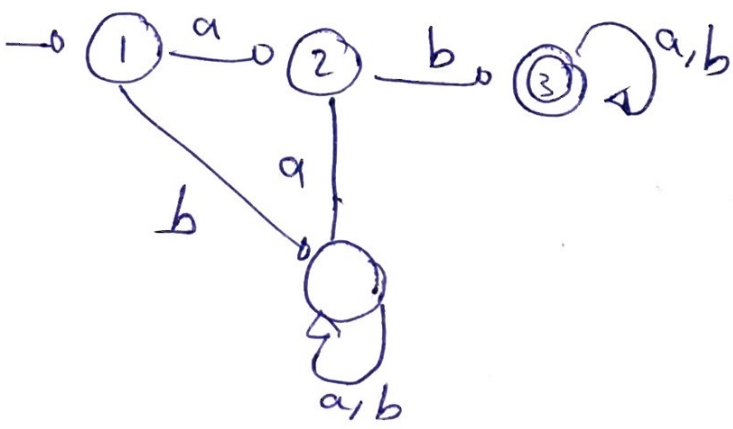


* Design DFA M for $L = \{a, ab\}$

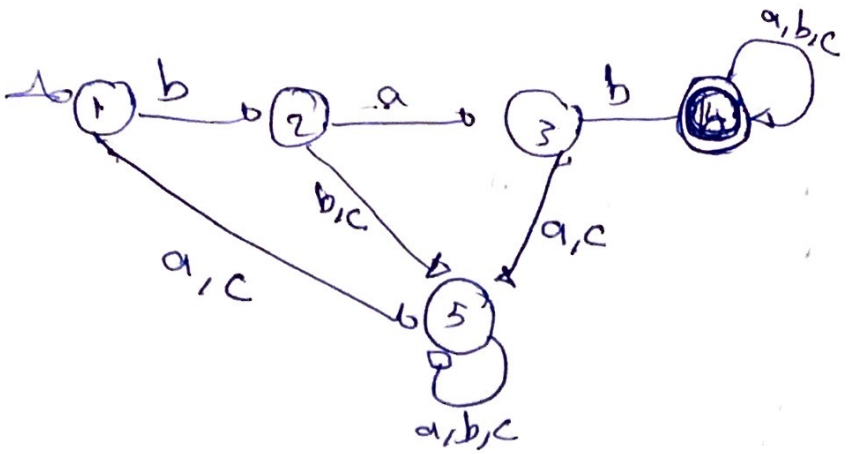


* Design DFA for $L = \{abw : w \in \Sigma^*, \Sigma = \{a,b\}^*\}$

abaa ✓
abbb ✓
abab ✓
ab ✗ ✗ ✗



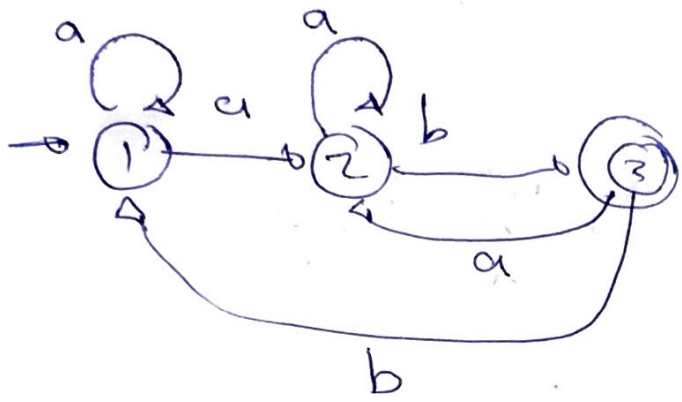
* Design DFA for $L = \{babw, w \in \Sigma^*, \Sigma = \{a,b,c\}\}^*$



babc ✓
baba ✓
babaaaa ✓
bab ✗✗✗

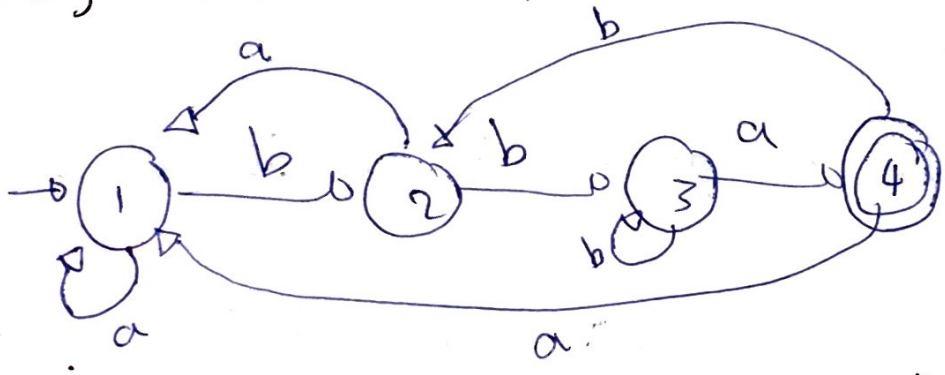
تاخذ كل الاحتمالات .

* Design DFA for $L = \{wab : w \in \Sigma^*, \Sigma = \{a,b\}\}$

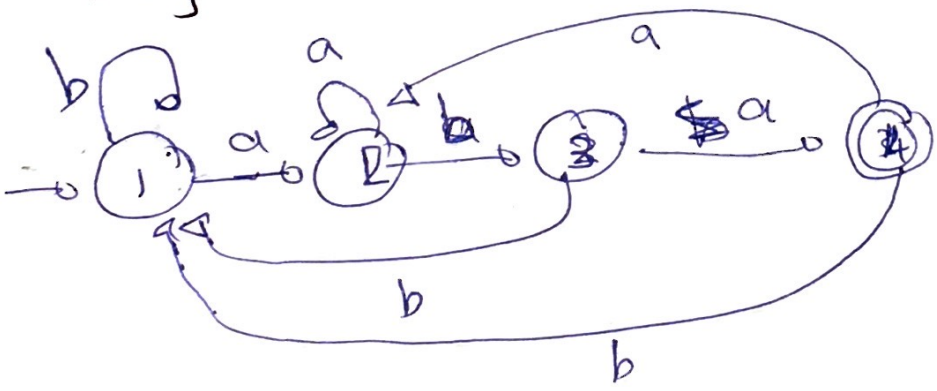


aaab
abab
aaaab
✗✗✗ab

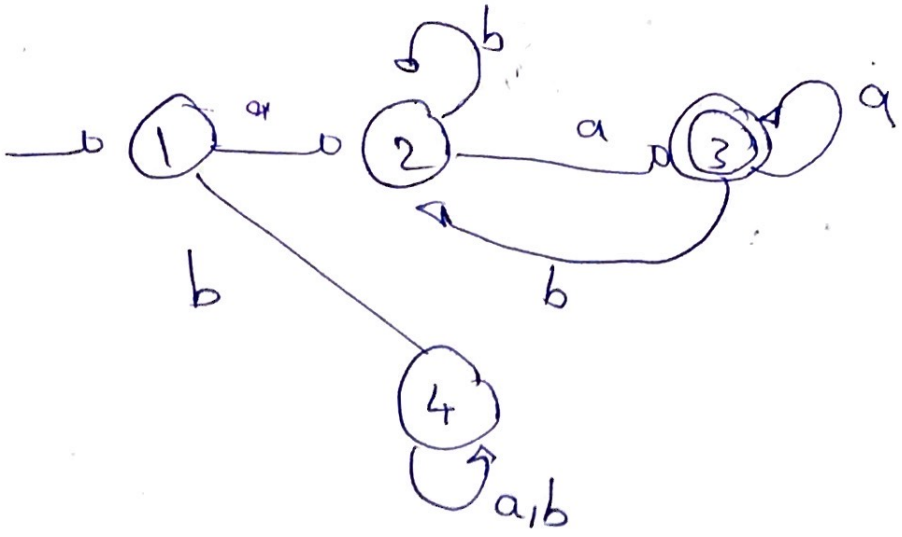
* Design DFA for $L = \{wbba, w \in \Sigma^*, \Sigma = \{a,b\}\}$



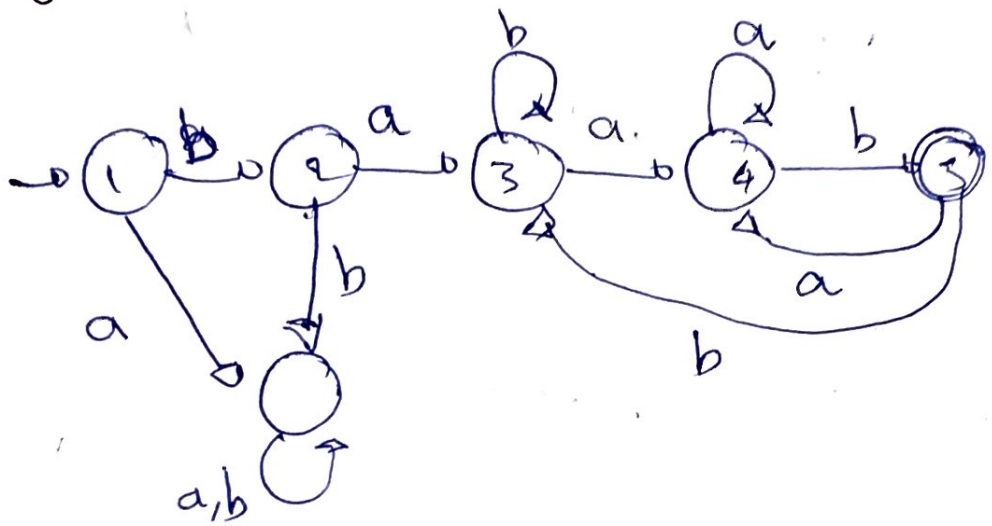
* Design DFA for $L = \{wabaa, w \in \Sigma^*, \Sigma = \{a,b\}\}$



* Design DFA $L = \{ awa , w \in \Sigma^* , \Sigma = \{a,b\}^* \}$

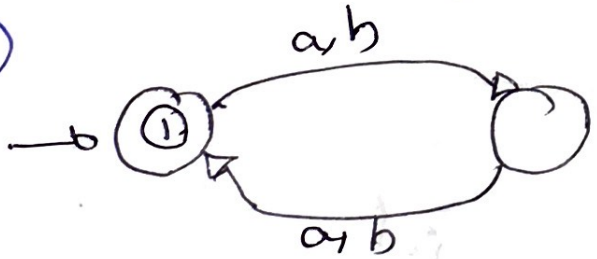


* Design DFA for $L = \{ bawab, w \in \Sigma , \Sigma = \{a,b\} \}$



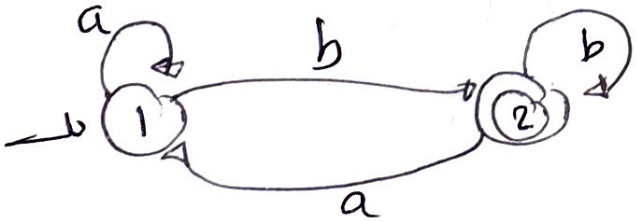
* what is the language accepted by this machine

(1)



$L = \{ w : w \in \{a,b\}^* , |w| \bmod 2 = 0 \}$

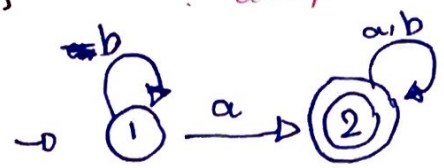
(2)



$L = \{ wb : w \in \{a,b\}^* \}$

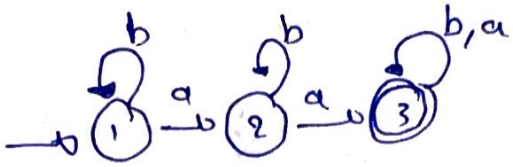
① Design DFA to L = {all strings in {a,b}* containing at least one a}

accept a واحدة أو أكثر تظهر مرة



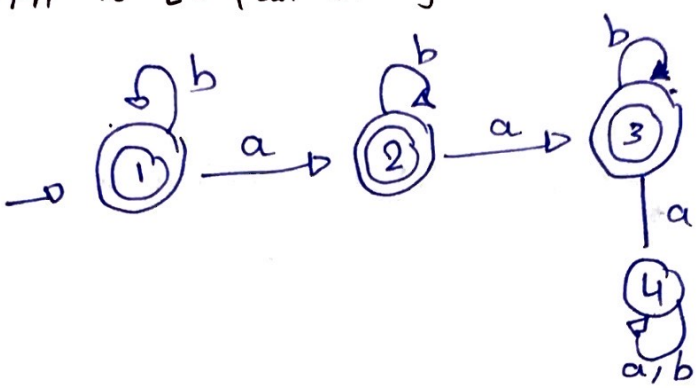
b a ab ✓
b b X
bab ✓
λ X

② Design DFA to L = {all strings in {a,b}* containing at least 2a}



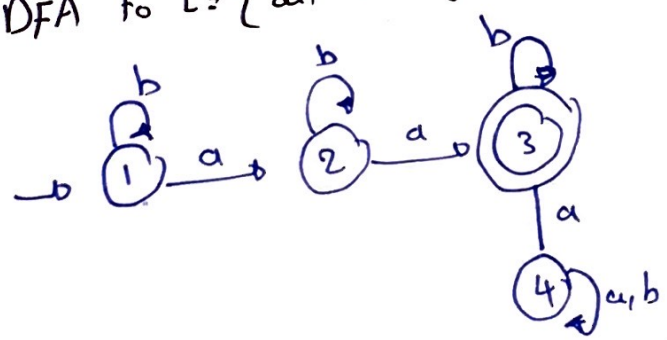
ab X
aacaab ✓
λ X

③ Design DFA to L = {all strings in {a,b}* containing at most 2 a's}



accept λ يعني
b ⇒ accept
bba ⇒ accept
bbaaab ⇒ reject

④ Design DFA to L = {all strings in {a,b}* containing exactly 2 a}

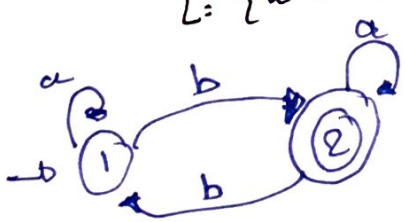


ab X
aab ✓
bbba X
aabaa X
baa ✓

⑤ Design DFA to L = {with odd numbers of b's and any number of a's}

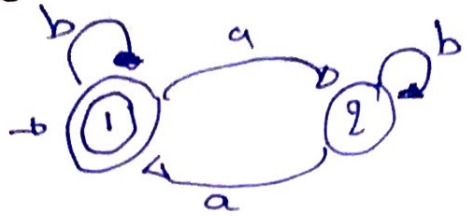
L = {w : w ∈ {a,b}* and number of b's is odd.

b's عدد فردي من



bba X
bbba ✓
bbbbaaa X
bbbbba ✓
λ X

⑥ Design DFA with $L = \{ w : w \in \{a,b\}^* \}$ and numbers of a's even }



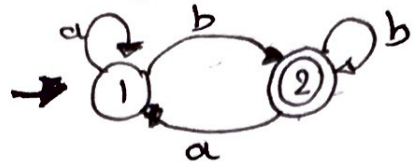
bba ✗
λ ✓
aab ✓
aaa bb ✗

even = زوجي، اذا اعتبرنا 0 زوجي

---

* what the language accepted by machine :

ج1



$L = \{ wb : w \in \{a,b\}^* \}$

w b
aaaa bbba b
Σ^*

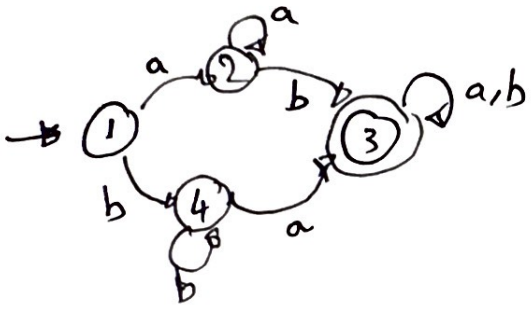
لازم تنتهي بـ b

ج2



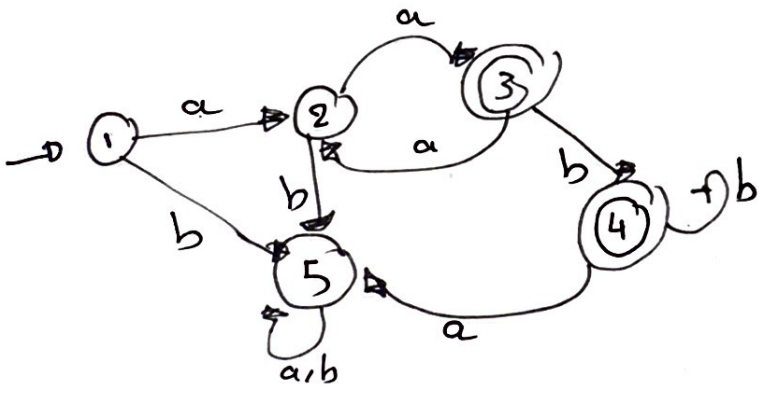
L = { all strings that contains at least 2 a }

ج3



L = { any string with an a and ab }

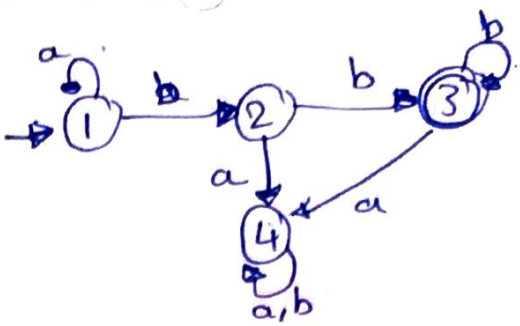
ج4



$L = \{ a^{2n} b^m : n \geq 1, m \geq 0 \}$

* Design DFA for :

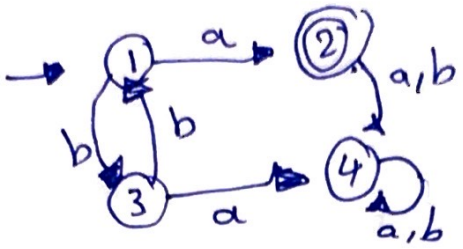
1- $L = \{ a^n b^m , n \geq 0 , m \geq 2 \}$



bb ✓

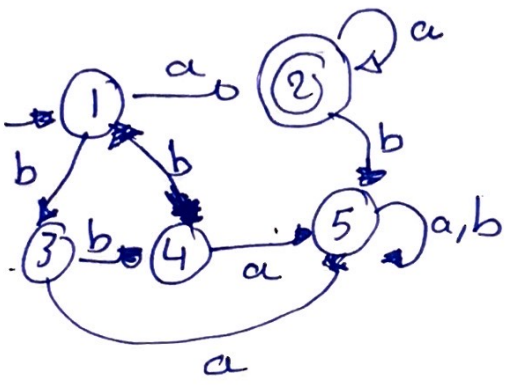
aaabb ✓
aabba X
لازم تكون مرتبة
يعني a بعدين b
X

2- $L = \{ (bb)^n a \ , n \geq 0 \}$



a ✓
bbba ✗
abbbb ✗

3- $L = \{ (bbb)^n a^m : n \geq 0 , m \geq 1 \}$



End of DFA

* NFA : Non-Deterministic Finite Automata.

اللي الفرق الـ DFA ما بنحط λ صبتجا على بعضها NFA ب دايماً بناخذها بعين الاعتبار

Formal Definition $N = (Q, \Sigma, \delta, q_0, F)$

δ تختلف عن DFA

$$\delta = Q \times \Sigma \cup \{\lambda\} \rightarrow P(Q)$$

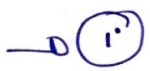
*** Case 1**



$\delta(i, a) = j$

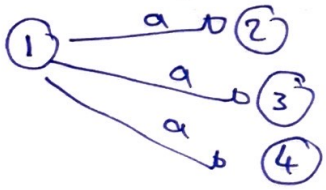
كل state بكون عليها λ بس ما بنكون شايفينها

*** Case 2**



$\delta(i, a) = \emptyset$

Case 3



$\delta(1, a) = \{2, 3, 4\}$

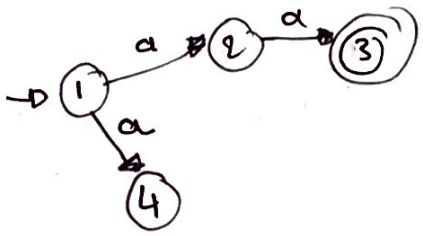
*** Case 4**



$\delta(i, \lambda) = j$

$\delta(i, \lambda) = \{i, j\}$

*Example 1

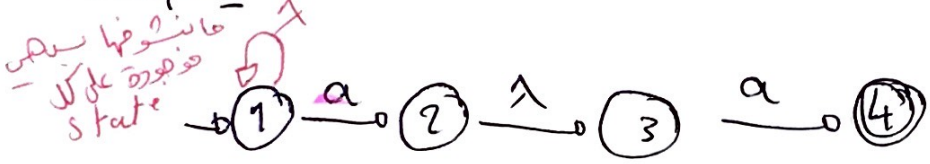


input	output
a	$\{2,4\} \cap F = \emptyset$ reject
aa	$3 \cap F \neq \emptyset$ accept
aaa	reject
λ	reject

$L(N) = \{aa\}$

$$L(DFA) = \{w : w \in \Sigma^* \text{ and } \delta^*(q_0, w) \cap F \neq \phi\}$$

* Example 2



input	output
λ	1 reject
a	2,3 reject
aa	4 accept
aaa	reject

$a\lambda a$

$L(N) = aa$

$\delta^*(1, aa) = \{4\}$

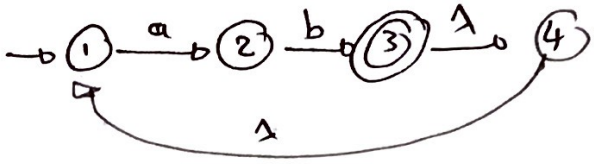
extended transition Function

*tracing rewrite $\lambda a \lambda a \lambda$?

$\{1\} \xrightarrow{a} \{2\} \xrightarrow{\lambda} \{2,3\} \xrightarrow{a} \{4\} \xrightarrow{\lambda} \{4\}$

- Dumber language

Exam Example [3]



input	output
λ	{1} reject
a	{2} reject
aa	{ } reject
ab	{1, 3, 4} accept ∈ Final state

ab λ λ
1, 3, 4 عشان وصلت

* f

*tracing :

$\delta^*(1, abab) \Rightarrow \lambda a \lambda b \lambda ab \lambda$

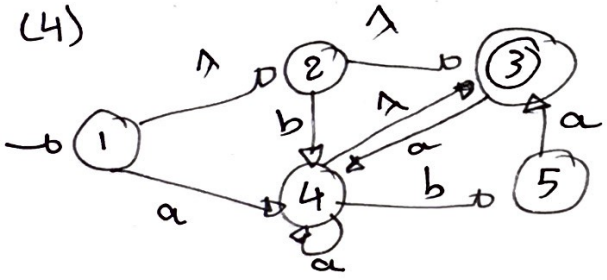
$\{1\} \xrightarrow{\lambda} \{1\} \xrightarrow{a} \{2\} \xrightarrow{b} \{3\} \xrightarrow{\lambda} \{4,3,1\} \xrightarrow{a} \{2\} \xrightarrow{\lambda} \{2\}$

$\xrightarrow{b} \{3\} \xrightarrow{\lambda} \{1,3,4\}$.

$L(N) = \{ab, abab, \ldots\}$

$L(N) = \{(ab)^n : n \geq 1\}$

Example (4)



input	output
λ	{1, 2, 3} accept
a	{3, 4} accept
b	{3, 4} accept
bb	{5} reject
aba	{3} accept

*trace of $\delta^*\{1, bb\} \Rightarrow bb \cdot \lambda b \lambda b \lambda$

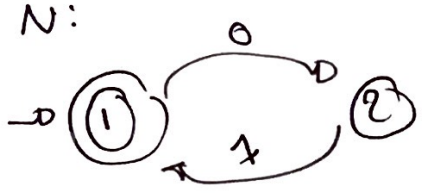
$\{1\} \xrightarrow{\lambda} \{1, 2, 3\} \xrightarrow{a} \{4\} \xrightarrow{\lambda} \{3, 4\} \xrightarrow{b} \{5\} \xrightarrow{\lambda} \{5\} \xrightarrow{a} \{3\} \xrightarrow{\lambda} \{3$

~~*trace of~~ $\delta^*(1, a) = \{3, 4\}$

$\delta^*(1, \lambda) = \{1, 3, 2\}$

Example (4)

N:



$L(N) = \{ 0^m \text{ where } m \geq 0 \}$

---

* Find L(N) for the following NFA :

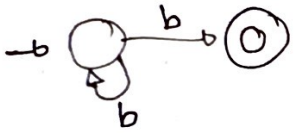
$L(N) = \{ w : w \in \Sigma^* , \delta^*(q_0, w) \in F \}$

(1)



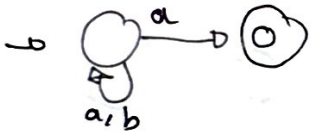
$L(N) = \{a\}$

(2)



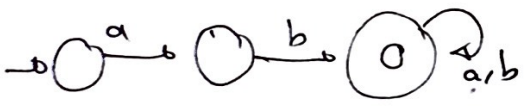
$L(N) = \{ b^n : n \geq 1 \}$

(3)



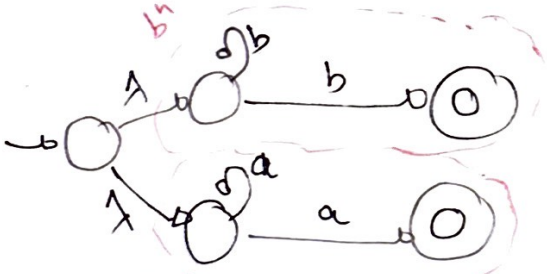
$L(N) = \{ wa : w \in \{a,b\}^* \}$

(4)



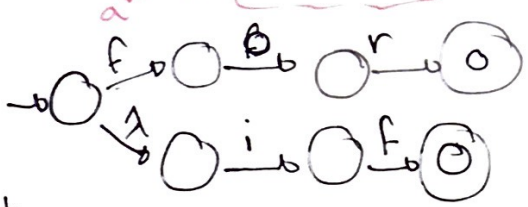
$L(N) = \{ abw : w \in \{a,b\}^* \}$

(5)



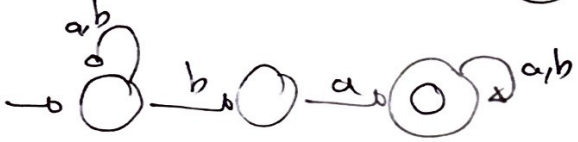
$L(N) = \{ a^n : n \geq 1 \} \cup \{ b^n : n \geq 1 \}$

(6)



$L(N) = \{ for, if \}$

(7)



$L(N) = \{ w_1 ba w_2 : w_1, w_2 \in \{a,b\}^* \}$

* Design NFA for following Languages

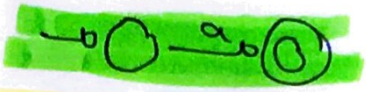
① $L = \{\}$



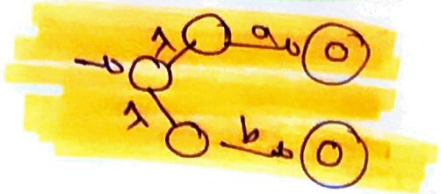
② $L = \{\lambda\}$



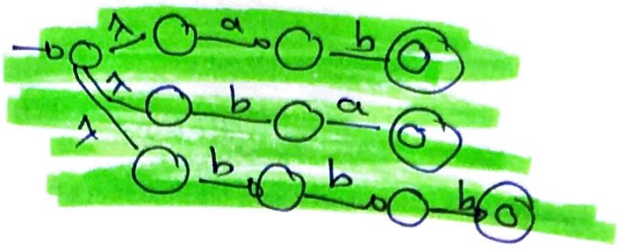
③ $L = \{a\}$



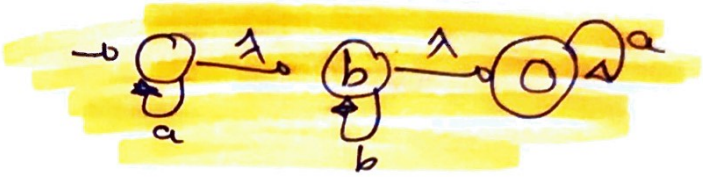
④ $L = \{a, b\}$



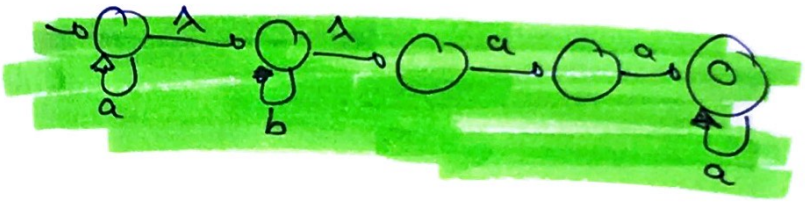
⑤ $L = \{ab, ba, bbb\}$



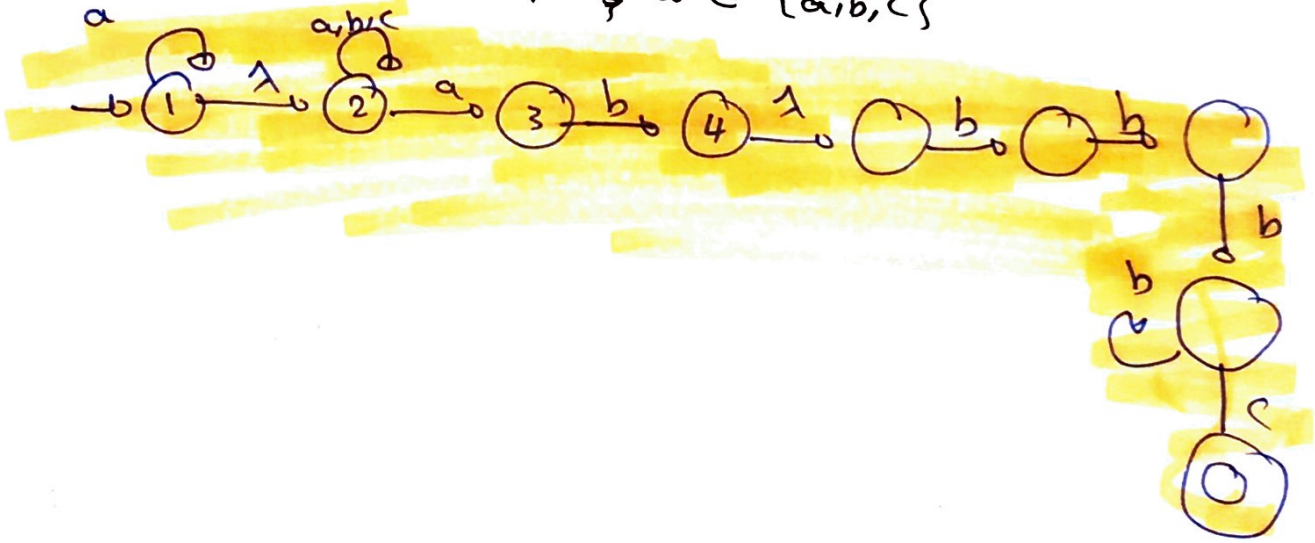
⑥ $L = \{a^n b^m a^k, n, m, k \geq 0\}$



⑦ $L = \{a^n b^m a^k, n, m \geq 0, k \geq 2\}$

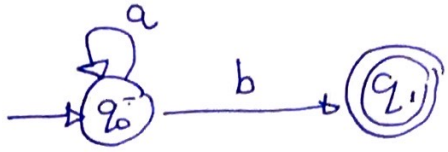


⑧ $L = \{a^n w ab b^m c : n \geq 0, m \geq 3\}$ $w \in \{a, b, c\}$



* Equivalence Machines

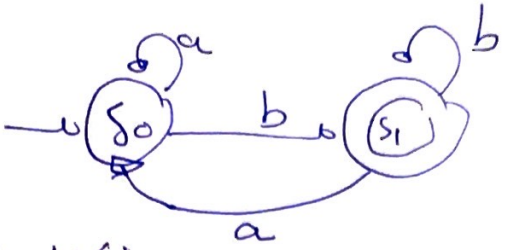
NFA:



$L(N) = \{wb : w \in \{a,b\}^*\}$

=

DFA:

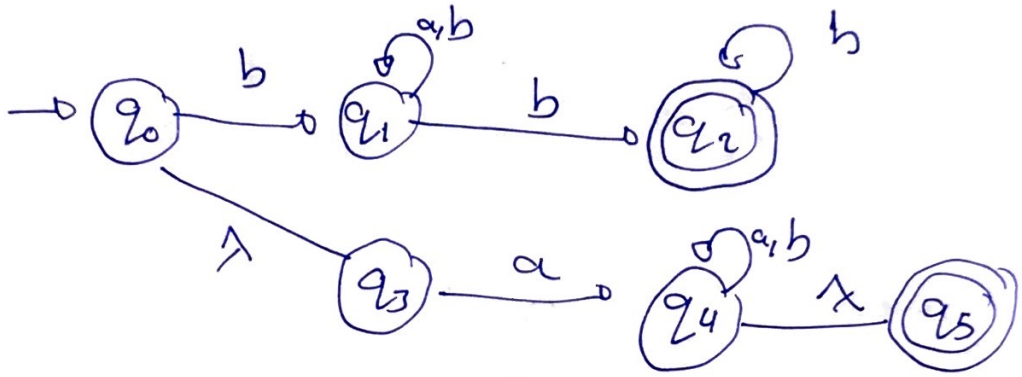


$L(D) = \{wb : w \in \{a,b\}^*\}$

$L(M) = L(M)$

نقدر اعمل اي NFA الى DFA والعكس

*example: convert NFA to DFA



input: NFA $= (Q, \Sigma, \delta, q_0, F)$

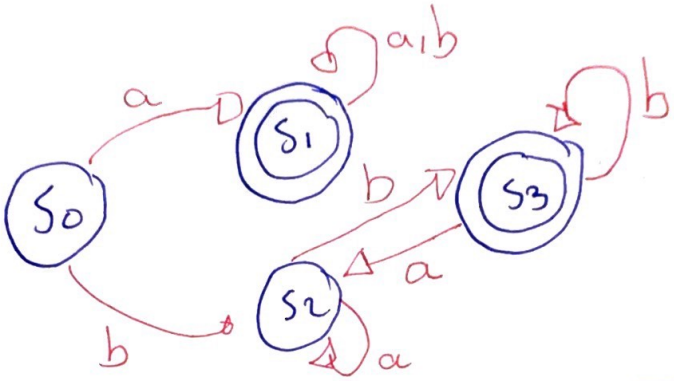
input DFA $= (\bar{Q}, \Sigma, \delta, \bar{q}_0, F)$

① compute q_0'

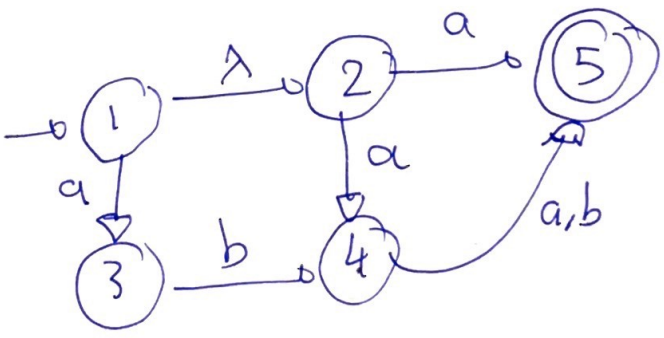
$\delta^*(q_0, \lambda) = \{q_0, \lambda\} = \bar{q}_0$

② compute δ_D

		a	b
−	S_0 (q_0, λ)	δ_1 (q_4, q_5)	S_2 (q_1)
+	S_1 (q_4, q_5)	δ_1 (q_4, q_5)	δ_1
	S_2 (q_1)	S_2	δ_3 (q_1, q_2)
+	S_3 (q_1, q_2)	S_2	S_3



Example 2: Convert NFA to DFA



① compute q_0 =

$$\delta^*(q_0, \lambda) = \delta^*(1, \lambda) = (1,2) = A$$

A جديد

* Regular Language

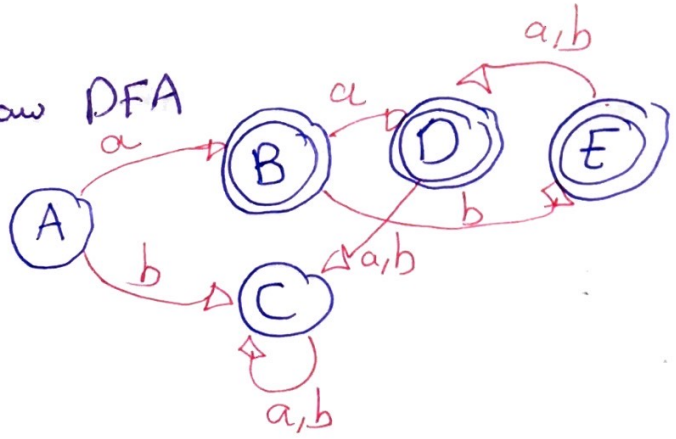
② Compute S_D:

		a	b
−	A (1,2)	$\delta^*(A,a)$ (3,4,5) B	$\delta^*(A,b)$ Φ C
+	B (3,4,5)	$\delta^*(B,a)$ {5} D	$\delta^*(B,b)$ {4,5} E
	C Φ	$\delta^*(C,a)$ { } C	$\delta^*(C,b)$ { } C
+	D {5}	$\delta^*(D,a)$ { } C	$\delta^*(D,b)$ { } C
+	E {4,5}	$\delta^*(E,a)$ {5} D	$\delta^*(E,b)$ {5} D

③ compute F_D

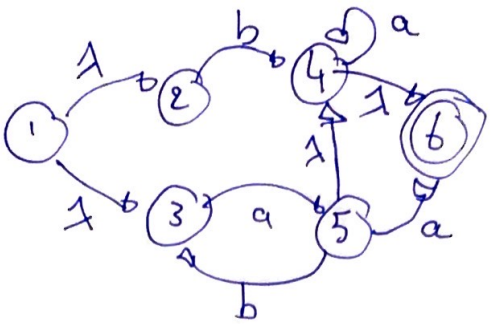
$F_D = (B, D, E)$

④ Draw DFA



* Regular Language

Example : convert NFA to DFA



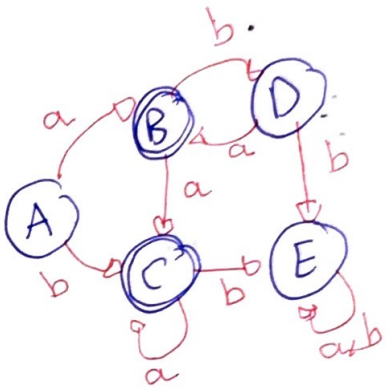
① compute q_0

$\delta^*(1,\lambda) = \{1,2,3\}$ → (A)

② compute SD

	a	b
A {1,2,3}	$\delta^*(A,a) = \{4,5,6\}$ B	$\delta^*(A,b) = \{4,6\}$ C
+ B {4,5,6}	$\delta^*(B,a) = \{4,6\}$ C	$\delta^*(B,b) = \{3\}$ D
+ C	$\delta^*(C,a) = \{4,6\}$ C	$\delta^*(C,b) = \{\}$ E
D {3}	$\delta^*(D,a) = \{4,5,6\}$ B	$\delta^*(D,b) = \{\}$ E
E {}	$\delta^*(E,a)$ E	$\delta^*(E,b)$ E

④ Draw DFA



③ compute Final : B, C

* Any language we can represent in

DFA ⟷ NFA ⟷ Regular Expression.

* Regular Expression:

$R = (a \oplus bc)^*$ — (⊕ = or, concatenation, * = kleene closure, * = 0, 1, 2, 3, ... ∞)

* example:

$L(R) = \{a, bc\}^*$

$= \{\underset{0}{\lambda}, \underset{1}{a}, \underset{2}{abc}, aa, abc, bca, bcbc, \underset{3}{aaa}, aabc, \dots bcbcbc\}$

* Def: let Σ be agiven alphabet

① primitive regular expression.

1. $\emptyset \Rightarrow L(\emptyset) = \{\}$
2. $\lambda \Rightarrow L(\lambda) = \{\lambda\}$
3. $a \Rightarrow L(a) = \{a\}$

② Regular Expression operators

if r1 and r2 are regular Expression then:-

1. $r_1 \oplus r_2$ (normal script) or $r1/r2$ (union)

$L(r1+r2) = L(r1) \cup L(r2)$

2. $r1.r2$ or $r1\,r2$

$L(r1.r2) = L(r1\,r2) = L(r1) \cdot L(r2)$

3. r^*

$L(r^*) = (L(r1))^*$

④ (r_1)

$L((r_1)) = L(r_1)$

⑤ r_1^+ superscript

$= r_1 r_1^* = r_1^* r_1$

⑥ $r_1?$ $= \lambda + r_1$

or

? optional

? 0, 1

⑦ $r_1\{5\}$

r_1 تتكرر 5 مرات

خاصة لغات البحة

⑧ $r_1\{1,3\}$ // $r_1\{,5\}$ // $r_1\{2,\}$

من 1 الى 3

Example: assume $\Sigma = \{a, b, c\}$

$r = (a + b.c)^* . (c + \emptyset)$

$L(r) = \{c, \{a, bc\}dc, bcc\}, aac, abc, bca, bcbc,$

② $aac, bcac, bcbcc, \ldots\}$

Example: Find $L(a^*.(a+b))$

$L(a^*.(a+b)) = L(a^*)\,L((a+b))$

$= L(a^*)\,L((a+b))$

$= \{a\}^*\,L(a+b)$

$\{\lambda, a, aa, aaa, \ldots\}\{a,b\} = \{a\}^*\,L(a) \cup L(b)$

$= \{a\}^*\,\{a,b\}.$

$= \{a, b, aa, ab, aaa, aab, aaaa, aaab, \ldots\}$

(bb + aa)+

Example 2

Find $L((a+b)^* ab)$

$= L((a+b)^* \cdot ab)$

$= L(a+b)^* \bullet L(ab)$

$= (L(a \cup b))^* \bullet L(a) \bullet L(b)$

$= (\{a\} \cup \{b\})^* \bullet \{a\}\{b\}$

$= \{a,b\}^* \{ab\}$

$= \{\lambda, ab, aa, ab, ba, bb, \ldots\} \{ab\}$

$= \{ab, aab, abab,$

$= \{wab : w \in \{a,b\}^*$

string های ab و ... ab

Example : Find $L(b.(a+b)^*)$

$= L(b) \bullet (L(a+b))^*$

$= \{b\} \bullet (L(a) \cup L(b))^*$

$= \{b\} \bullet \{a,b\}^*$

$= \{b\} \bullet \{\lambda, a, b, ab, aab, \ldots\}$

$= \{b\} \{a,b\}^*$

$= \{bw, w \in \{a,b\}^*\}$

Example Find L(r)

① $r = (a+b)^* (a+bb)$

$= L(r) =$

$= \{wa : w \in \{a,b\}^*\} \cup \{wbb, w \in \{a,b\}^*\}$

② $r = (a+b)(a+b)$
$\{a,b\}\ \{a \cup b\}$

$L(r) = \{aa, ab, ba, bb\}$

③ $r = (a+b)^* aa (a+b)^*$

$L(r) = \{w_1 aa w_2 : w_1, w_2 \in \{a,b\}^*\}$

---

* write regular expression for the following Languages

① $L = \Sigma^*$, $\Sigma = \{a,b,c\}$

$r = (a+b+c)^*$

② $L = \{a^n : n \geq 2\}$ تقريبي

$r = aaa^*$ or aa^+

③ $L = \{a^{2n} : n \geq 0\}$

$r = aa^*$

④ $L = \{a^{3n} : n \geq 0\}$

$r = aaa^*$

⑤ $L = \{a^{3n}, n \geq 2\}$

$r = aaaaaa(aaa)^*$ or $(aaa)(aaa)^+$

(6) $L = \{ a^{2n} b^m c^k : n, m, k \geq 1 \}$

$r = (aa)(aa^*)\ bb^*\ cc^*$ or $(aa)^+ b^+ c^+$

(7) $L = \{ w_1 010 w_2 : w_1, w_2 \in \{0,1,2\}^* \}$

$r = (0+1+2)^*\ 010\ (0+1+2)^*$

(8) $L = \{ awb : w \in \{a,b,c\}^* \}$

$r = a(a+b+c)^* b$

(9) L = C++ comments

① $r = '//'\ a^*\ '\backslash n'$ → single comment

② $r = '/'\ '*'\ a^*\ '*'\ '/'$

* $r = '/'\ '/'\ a^*\ '\backslash n' + '/'\ '*'\ a^*\ '*'\ '/'$

(10) C++ ID - letter/digits

$r_{Digits} = 0+1+2+3+\cdots 9 = 0-9$

$r_{letters} = A+B+C+\cdots Z + a+b+c\cdots z$

$= A-Z\ a-Z$

$r_{ID\ C++} = ('_' + r_{letters})('_' + r_{letters} + r_{Digits})$

(11) C++ Integers. $r_{Digit} = 0-9$

$r_{integers} = ('+' + '-')\ r_{Digits}^+$

(3)

(12) L = Regwords .

Regwords = if + else + for + while

(13) L = assignments operators .

L = " = ' + '+ = ' + ' - = ' + ' / = ' + ' % = '

+ ' * = '

(14) Strings

L = " r character* " + r character* '

(15) L = {ab, ba, abaa, bbb}

r = ab + ba + abaa + bbb

r = ab(λ + aa) + b(a + bb)

or r = (b + aba)a + (a + bb)b

25

(16) Σ = {a, b}

L = all strings with at least one a

r = a + aa + aa + abb + ---- a^n

r = $(a+b)^* a (a+b)^*$

(17) L = all strings with exactly one b

r = $a^* b a^*$

r = (bb + aa)+

* Regular Language

DFA NFA RegEx

L(r) = regular Language.

regular expression.

* Regular expression

① convert r to NFA

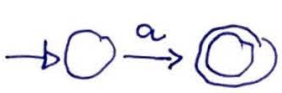
② convert NFA DFA to RegEx

① primitive Reg Exp.

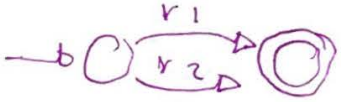
NFA / DFA

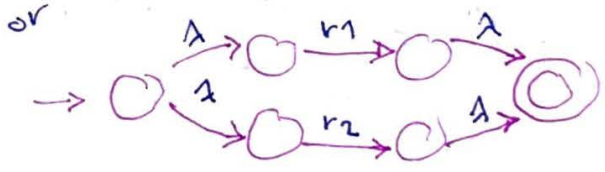
(a) $\emptyset$ 

(b) λ 

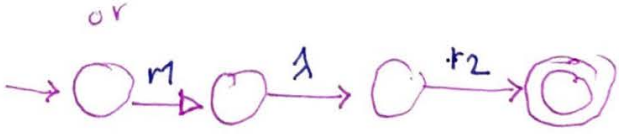
(c) $a \in \Sigma$ 

② RegEx operators, r1, r2 are Reg ex

a- $r_1 + r_2$ 

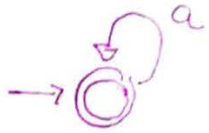
or 

b- $r_1 \cdot r_2$ 

or 

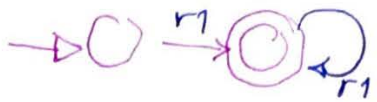
C- r_1^*

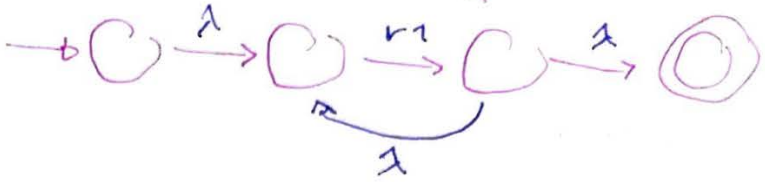
ex: ① a^*



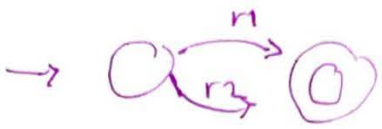
② $(a+b)^*$

D- $r+$



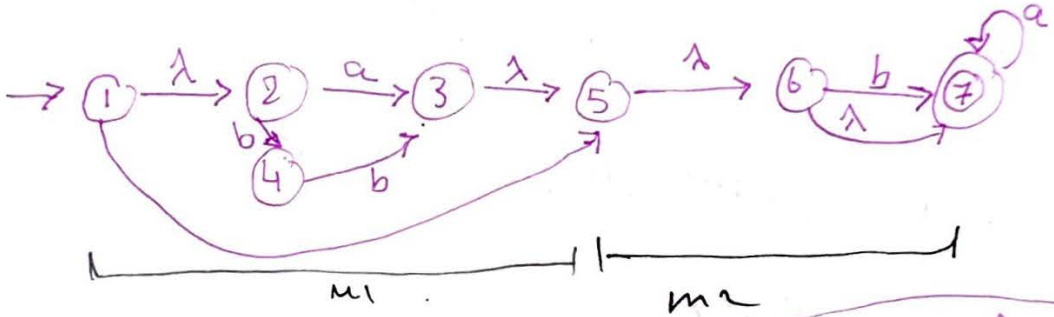


e $r1? = r1 + r2$

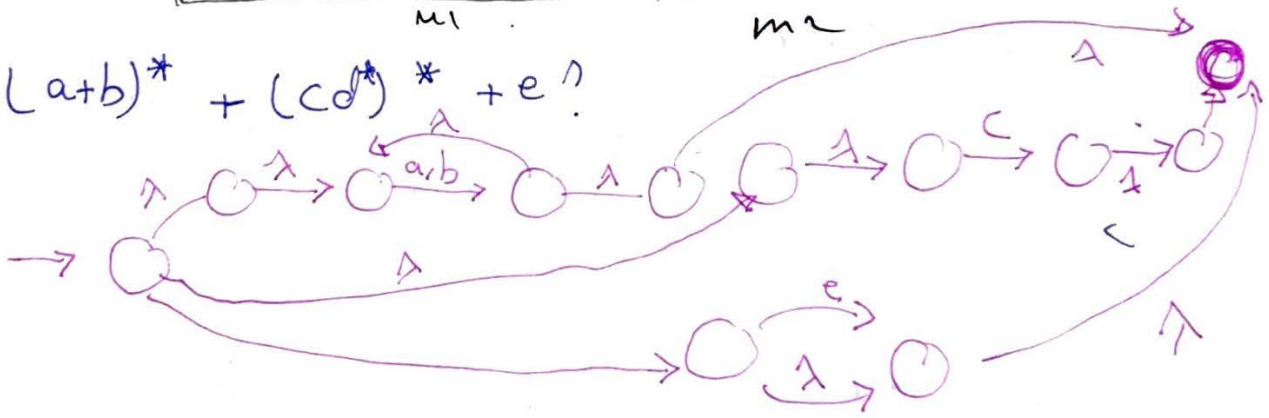


Example : find NFA that accept L(r)?

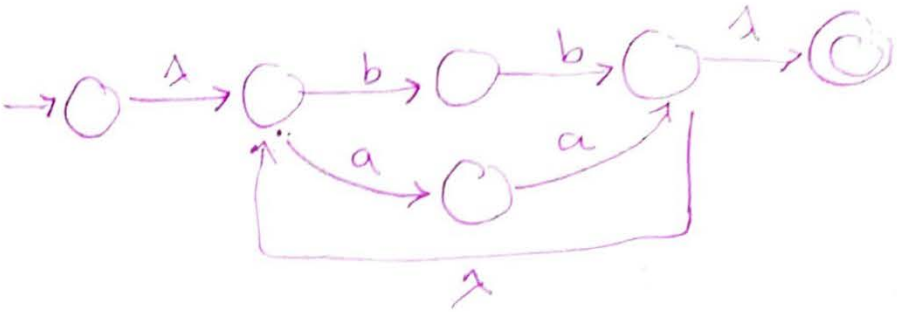
① $r = \underbrace{(a+bb)^*}_{M1}\ \underbrace{(ba^* + 1)}_{M2}$



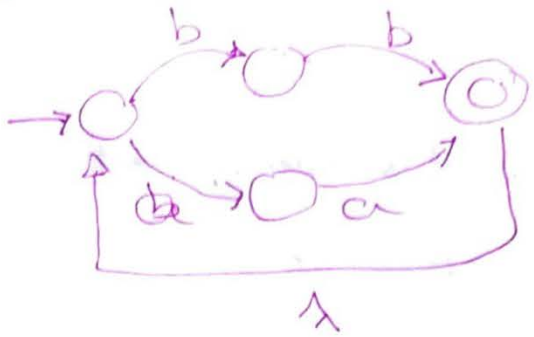
② $(a+b)^* + (cd^*)^* + e?$



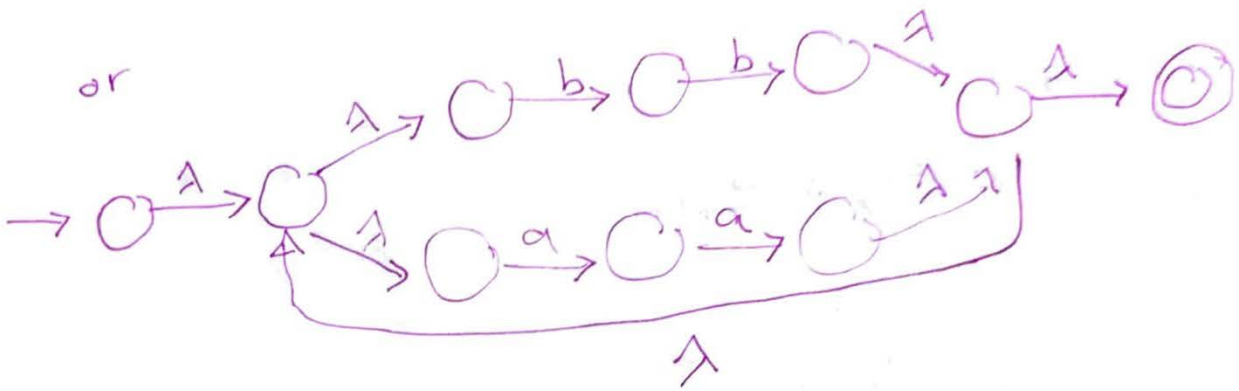
③ r: (bb + aa)+



or

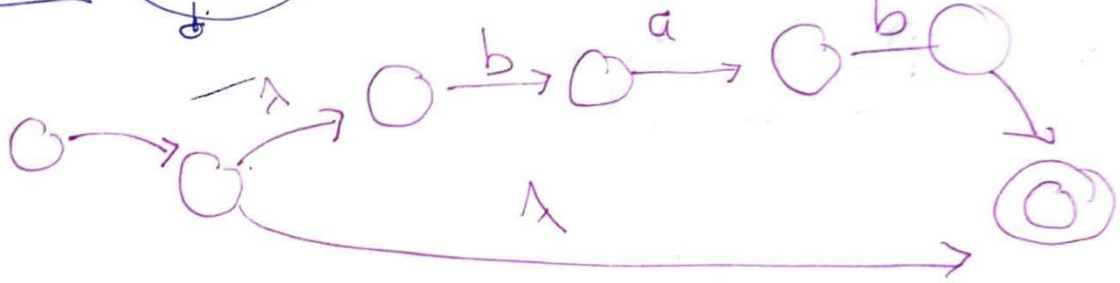


or



④ (bb + aa)+ (bab)*

الى كل
الا فوق



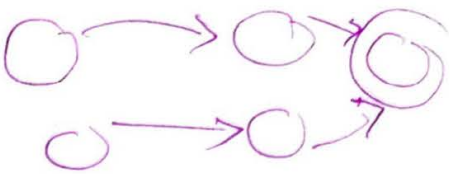
L(r) is regular Language:

① convert regEx to NFA

② Convert FA(NFA)/DFA) to RegEx

Conversion by elimination FA → RE

* Cases

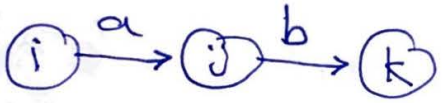


حيث كل الـ states ما عدا

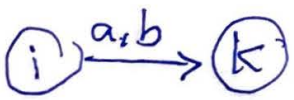


Case #1

ex:

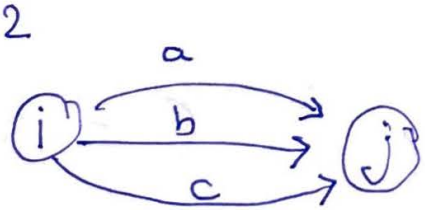


remove state j

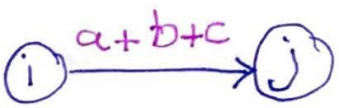


Case #2

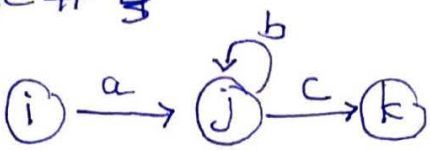
ex



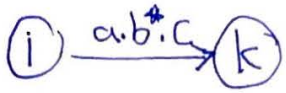
parallel الـ or يصير



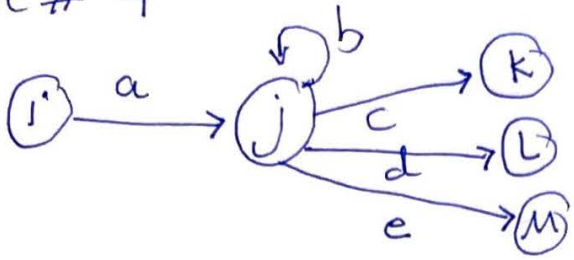
Case # 3



remove j



Case # 4

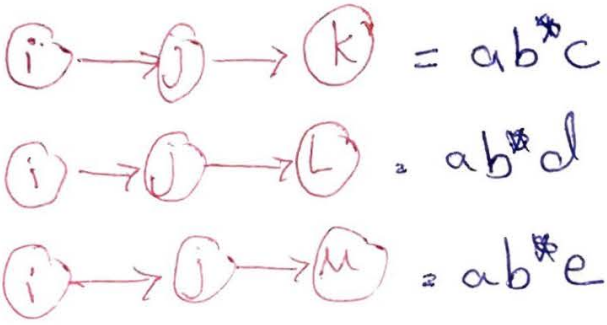


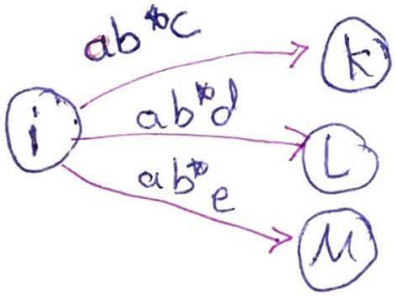
ملاحظة
الـ loop لا تحسب مع الـ in و الـ out

remove j

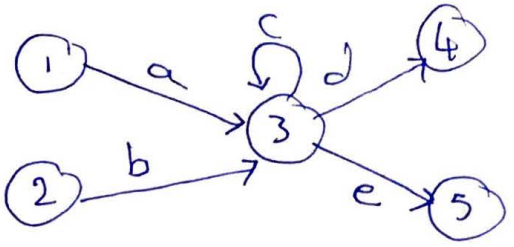
in = 1 out = 3

paths = in*out = 1*3 = 3





Case #5



* remove 3

in = 2 out = 2

#paths = 4



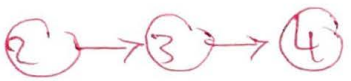
ac*d



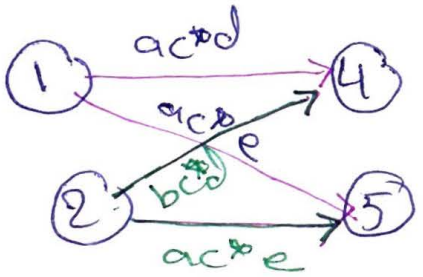
bc*d



ac*e



bc*e



Case # 6

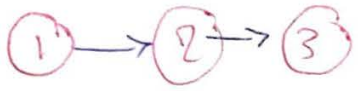


remove State 2.

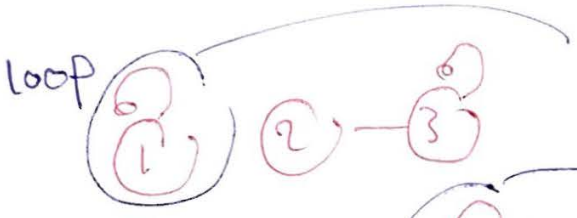
paths = in=2 * out=2 = 4



bd^*e



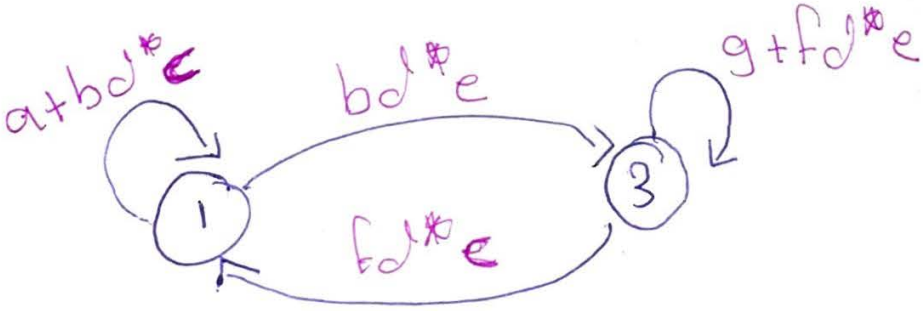
fd^*e



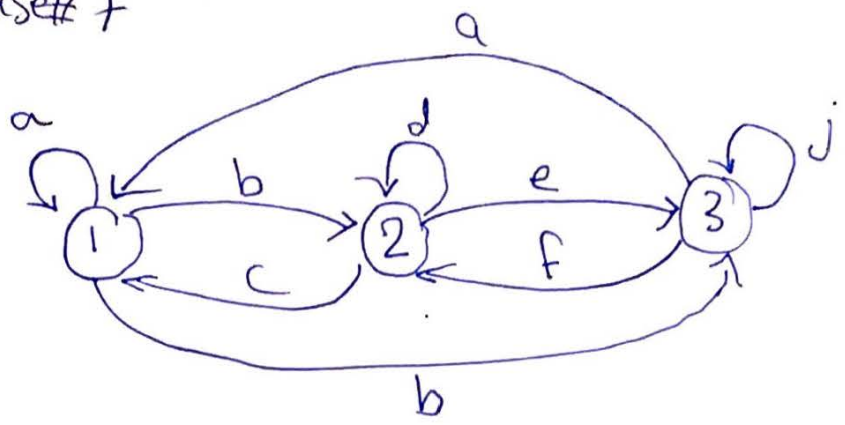
~~$a+bd^*c$~~ $a+bd^*c$

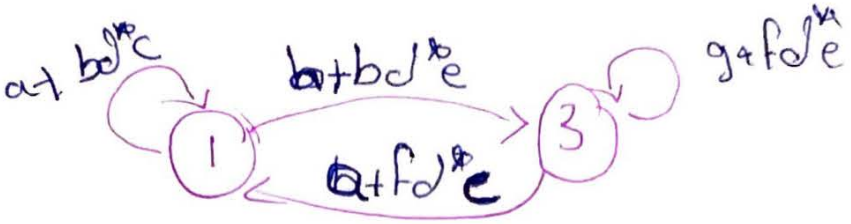


$g+fd^*e$

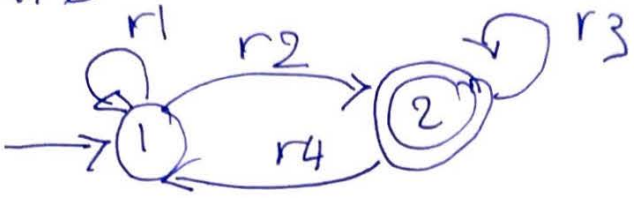


Case# 7



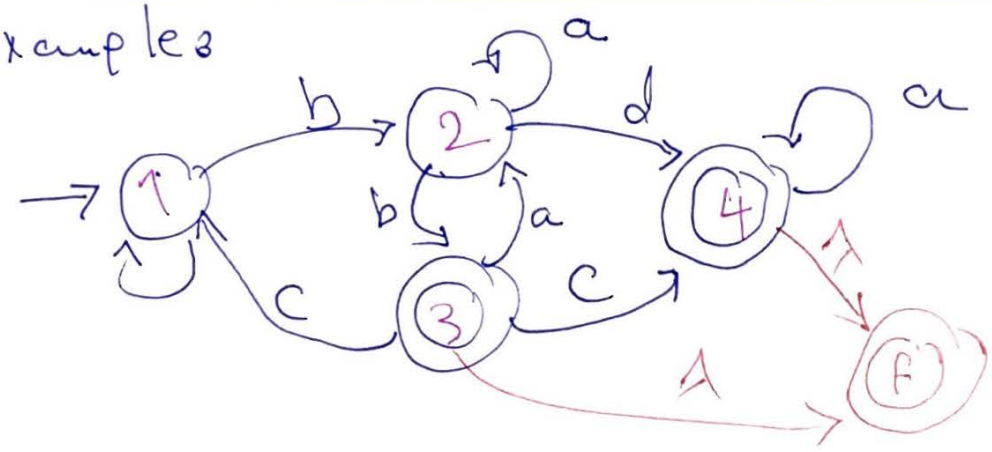


Case #8



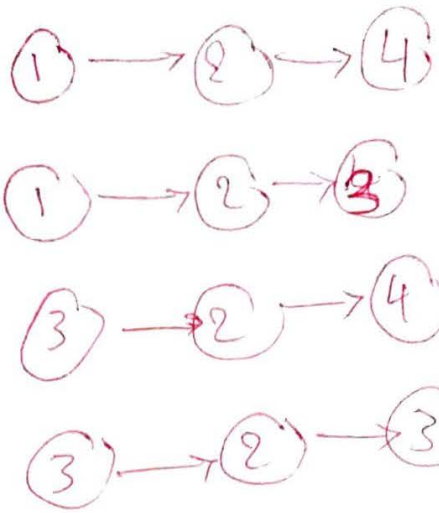
$$r = r1^{*} r2 (r3 + r4\, r1^{*} r2)^{*}$$

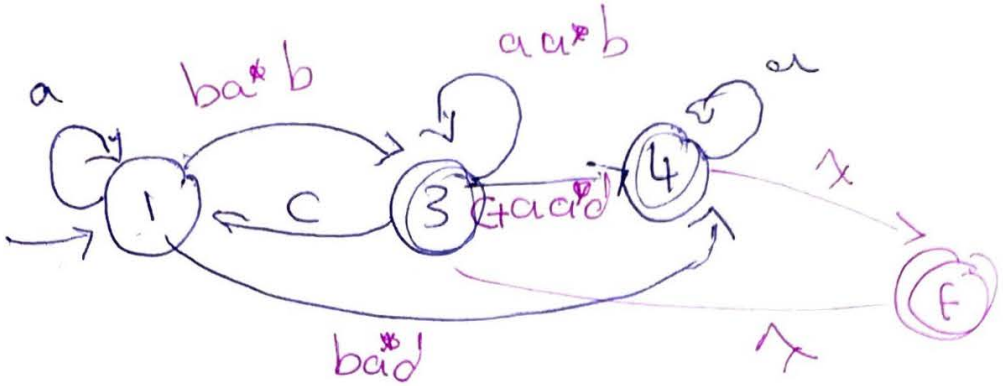
Examples



① remove state 2

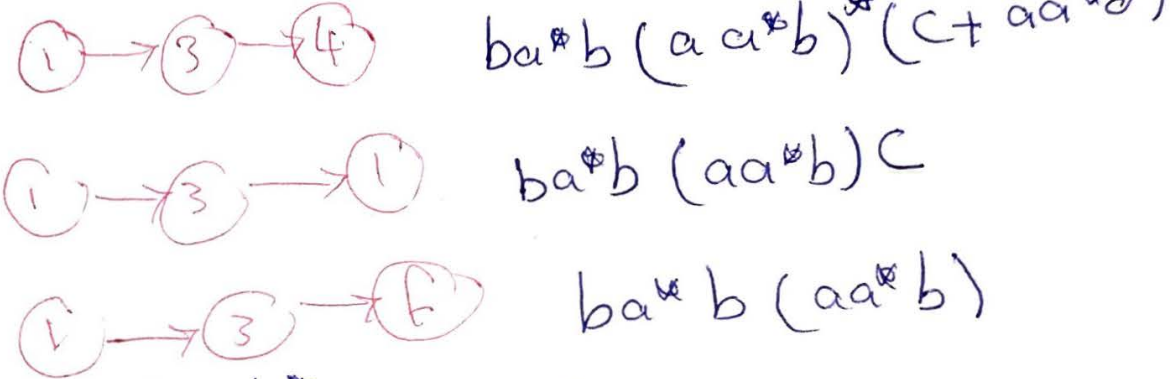
paths = in = 2 * out = 2 = 4



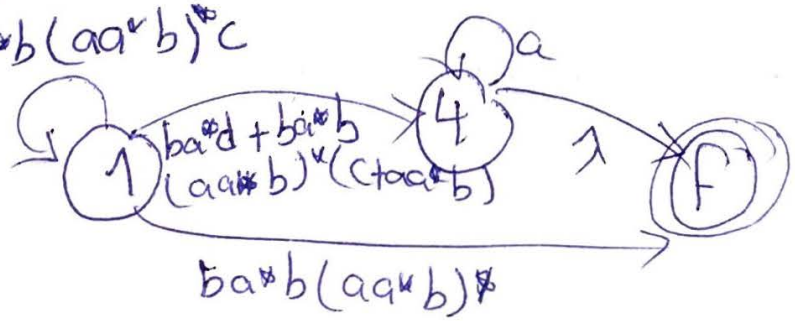


remove state (3)

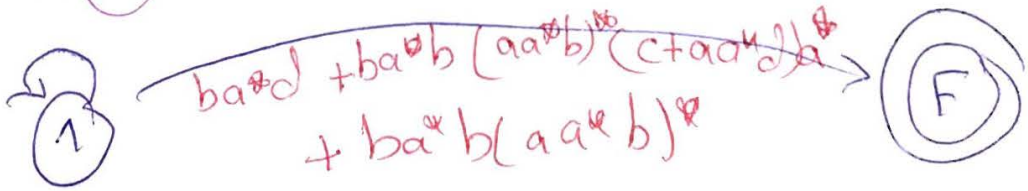
paths: in: 1 out: 3



$a + ba^*b(aa^*b)^*c$

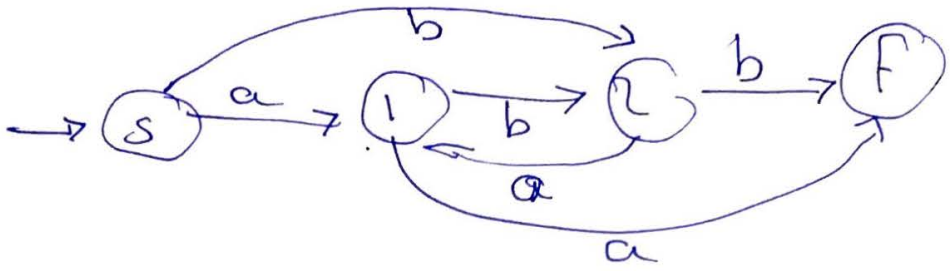


remove (4)



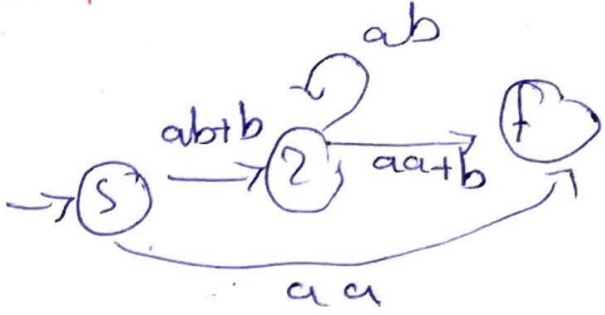
→ FA / DFA / NFA to RE

ex: convert NFA to RE



remove (1)

in = 2 out = 2 path = 4



S → 1 → 2 ab

S → 1 → f aa

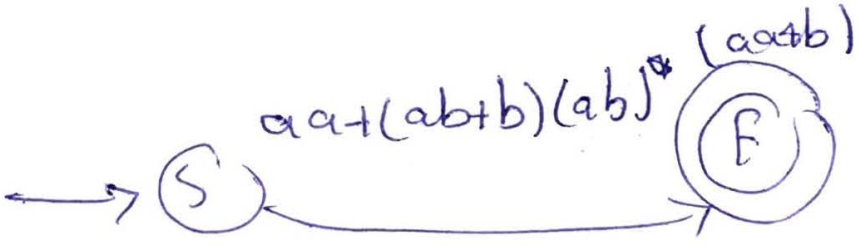
2 → 1 → f ab

2 → 1 → f aa

remove (2)

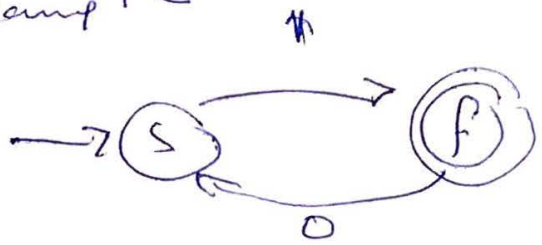
in 1 out = 1 paths = 1

S → 2 → F : (ab+b)(ab)*(aa+b)

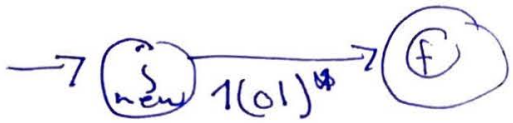


r : aa + (ab+b)(ab)*(aa+b)

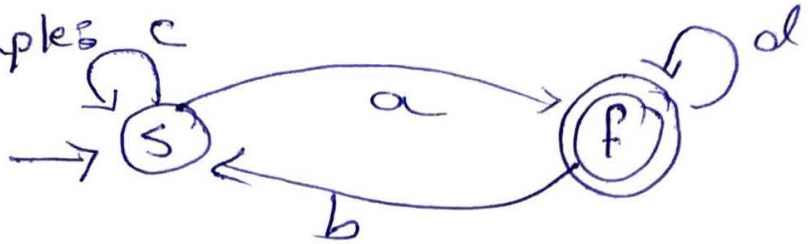
Example



Answer

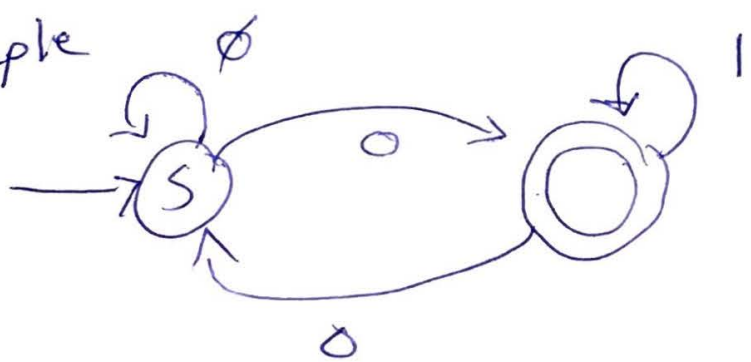


Example



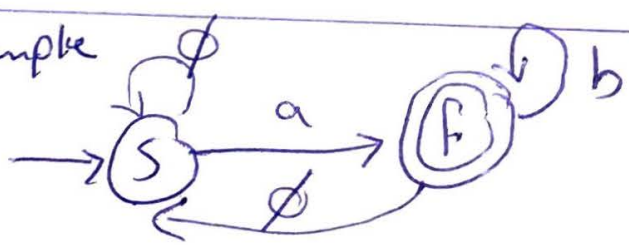
$r: c^*a(d + bc^*a)^*$

Example



$r = 0(1+00)^*$

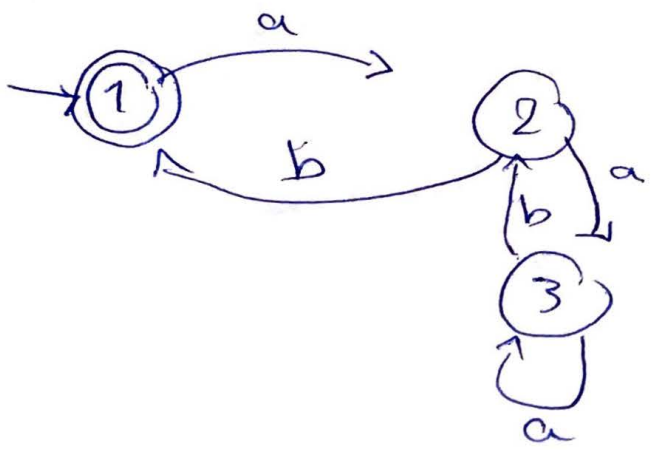
Example



$r = a(b)^*$

$r = ab^*$

Example



$r = a(aa^*b)^*b)^*$

Final

Regular Grammar

topic 3

*def $G = (P, \Sigma \text{ or } T, V, S)$

- P: production rules
- Σ or T: Terminals
- V: variables
- S: start variable

$P \rightarrow (\Sigma \cup V)^*$

$\downarrow$

$\in V$

*Example :

G: 1. $A \rightarrow aA$
2. $A \rightarrow bB$
3. $B \rightarrow C$

$\Sigma = \{a, b, c\} = T$

$V = NT(A, B)$

~~Not terminals~~

$S = A$

start

- Derivation of string aabc

$A \overset{*}{\Rightarrow} aabc$

$A \overset{(1)}{\Rightarrow} aA$

$A \overset{(1)}{\Rightarrow} aaA$

$A \overset{(2)}{\Rightarrow} aabB$

$B \overset{(3)}{\Rightarrow} aabc$

rule 1 تتكرر 2

rule 1 تتكرر 1 مرة

rule 2 تتكرر 1 مرة

* <sentence> → <NP><Predicate>

Example 2 :

G : 1. $S \to aSb$

2. $S \to \lambda$

* Derivation of aabb

$S \overset{*}{\Rightarrow} aabb$

$S \overset{(1)}{\Rightarrow} aSb$

$S \overset{(1)}{\Rightarrow} aaSbb$

$S \overset{(2)}{\Rightarrow} aabb$ ✓

Derivation of ab

$S \overset{*}{\Rightarrow} ab$

$S \overset{(1)}{\Rightarrow} aSb$

$S \overset{(2)}{\Rightarrow} ab$ ✓

$$L(G) = \{\lambda, ab, aabb, aaabbb, \dots\}$$

$$= \{a^n b^n : n \geq 0\}$$

Example 3 :

G : 1. $S \to (S)S$

2. $S \to \lambda$

right لازم لكل left

① Derivation of $(()())()$

$S \overset{*}{\Rightarrow} (()())()$

$S \overset{(1)}{\Rightarrow} (S)S$

$S \overset{(1)}{\Rightarrow} ((S)S)S$

$S \overset{(1)}{\Rightarrow} ((S)(S)S)S$

$S \overset{(2)}{\Rightarrow} (()(S)S)S$

$S \overset{(2)}{\Rightarrow} (()()S)S$

$S \overset{(2)}{\Rightarrow} (()())S$

$S \overset{(1)}{\Rightarrow} (()())(S)S$

$S \overset{(2)}{\Rightarrow} (()())()S$

$S \overset{(2)}{\Rightarrow} (()())()$ ✓

② Derivation $()(())$

$S \overset{*}{\Rightarrow} ()(())$

$S \overset{(1)}{\Rightarrow} (S)S$

$S \overset{(2)}{\Rightarrow} ()S$

$S \overset{(1)}{\Rightarrow} ()(S)S$

$S \overset{(1)}{\Rightarrow} ()((S)S)S$

$S \overset{(2)}{\Rightarrow} ()(()S)S$

$S \overset{(2)}{\Rightarrow} ()(())S$

$S \overset{(2)}{\Rightarrow} ()(())$ ✓

Example 4 :

Regular Grammar
RegEx

$G: 1. S \to aS$
$2. S \to \lambda$

① Derivation of aaa

$S \overset{*}{\Rightarrow} aaa$
$S \overset{(1)}{\Rightarrow} aS$
$S \overset{(1)}{\Rightarrow} aaS$
$S \overset{(1)}{\Rightarrow} aaaS$
$S \overset{(2)}{\Rightarrow} aaa$ ✓

② Derivation of aaaa

$S \overset{*}{\Rightarrow} aaaa$
$S \overset{(1)}{\Rightarrow} aS$
$S \overset{(1)}{\Rightarrow} aaS$
$S \overset{(1)}{\Rightarrow} aaaS$
$S \overset{(1)}{\Rightarrow} aaaaS$
$S \overset{(2)}{\Rightarrow} aaaa$ ✓

$L(G) = \{a^n : n \geq 0\}$

RegEx : $r = a^*$

Example 5 :

$G: 1. S \to SS$
$2. S \to a$
$3. S \to \lambda$

④ Derivation of λ :

$S \overset{*}{\Rightarrow} \lambda$
$S \overset{(3)}{\Rightarrow} \lambda$

② Derivation of aa

$S \overset{*}{\Rightarrow} aa$
$S \overset{(1)}{\Rightarrow} SS$
$S \overset{(2)}{\Rightarrow} aS$
$S \overset{(2)}{\Rightarrow} aa$

③ Derivation of a

① $S \overset{*}{\Rightarrow} a$
$S \overset{(2)}{\Rightarrow} a$

② $S \overset{*}{\Rightarrow} a$
$S \overset{(1)}{\Rightarrow} SS$
$S \overset{(2)}{\Rightarrow} aS$
$S \overset{(3)}{\Rightarrow} a$ ✓

$L(G) = \{a^n : n \geq 0\}$

Example 6

Regular Grammar.

G: 1. $S \to aS$
2. $S \to bS$
3. $S \to a$
4. $S \to b$

① Derivation of abbaa

$S \Rightarrow abbaa$

$S \overset{(1)}{\Rightarrow} aS$

$S \overset{(2)}{\Rightarrow} abS$

$S \overset{(2)}{\Rightarrow} abbS$

$S \overset{(3)}{\Rightarrow} abbaS$

$S \overset{(3)}{\Rightarrow} abbaa$ ✓

② Derivation of b

$S \Rightarrow b$

$S \overset{(4)}{\Rightarrow} b$ ✓

$L(G) = \{a, b, aa, ab, ba, \dots\}$

$L(G) = \{w : w \in \{a,b\}^+\}$

RegEx: $r = (a+b)^+$

Example 7

G: 1. $S \to X$
2. $S \to y$
3. $X \to \lambda$
4. $y \to ay$
5. $y \to by$
6. $y \to a$
7. $y \to b$

هون عنوع نبلش الا بـ S

هيز إذا حكينا $X \Rightarrow \lambda$ إنو نغلل لازم نحكي $S \Rightarrow \lambda$

$S \overset{(1)}{\Rightarrow} X$

$S \overset{(3)}{\Rightarrow} y$

الحل

$\Longrightarrow$

(4)

(5)

① Derivation of λ

$S \overset{*}{\Rightarrow} \lambda$
$S \overset{(1)}{\Rightarrow} X$
$S \overset{(3)}{\Rightarrow} \lambda$ ✓

② Derivation of baba

$S \overset{*}{\Rightarrow} baba$
$S \overset{(2)}{\Rightarrow} yy$
$S \overset{(5)}{\Rightarrow} by$
$S \overset{(4)}{\Rightarrow} bay$
$S \overset{(5)}{\Rightarrow} baby$
$S \overset{(6)}{\Rightarrow} baba$ ✓

$L(G) = \{w : w \in T^*\}$

$L(G) = (a+b)^*$

~~Example~~

Derivation Order

① left most derivation

$(\overset{}{\underset{Lm}{\Rightarrow}})$ نختار أقصى اليسار

Example 1

G: 1. $S \to AB$
2. $A \to aaA$
3. $A \to \lambda$
4. $B \to Bb$
5. $B \to \lambda$

$S \overset{*}{\underset{Lm}{\Rightarrow}} aab$

نختار $S \to AB$ لأنها أقصى اليسار، لذلك A أقصى اليسار

$S \overset{(1)}{\Rightarrow} AB$ هون بعوض A بـ λ
$S \overset{(2)}{\Rightarrow} aaAB$
$S \overset{(3)}{\Rightarrow} aaB$
$S \overset{(4)}{\Rightarrow} aaBb$
$S \overset{(5)}{\Rightarrow} aab$ ✓

② right most derivation

$(\Rightarrow_{rm})$ نختار أقصى اليمين

Example

G: 1. $S \to AB$
2. $A \to aaA$
3. $A \to \lambda$
4. $B \to Bb$
5. $B \to \lambda$

$S \overset{*}{\underset{rm}{\Rightarrow}} aab$

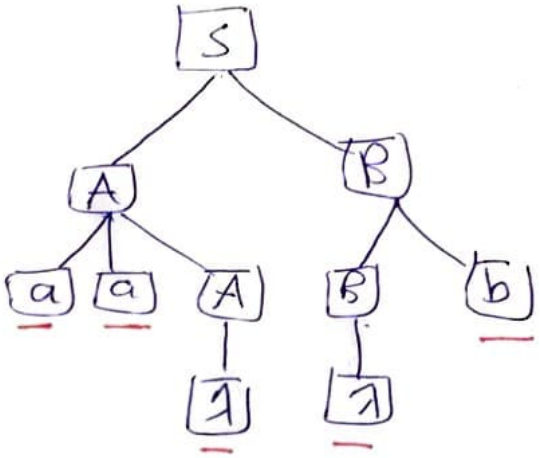
$S \overset{(1)}{\Rightarrow} AB$ هون بختار B أقصى اليمين
$S \overset{(4)}{\Rightarrow} ABb$ هون كمان بختار B أقصى اليمين
$S \overset{(5)}{\Rightarrow} Ab$
$S \overset{(2)}{\Rightarrow} aaAb$
$S \overset{(3)}{\Rightarrow} aab$ ✓

*Derivation Trees:

Root is Start NT

example 3(1)

G_3 1. $S \to AB$
2. $A \to aaA$
3. $A \to \lambda$
4. $B \to Bb$
5. $B \to \lambda$



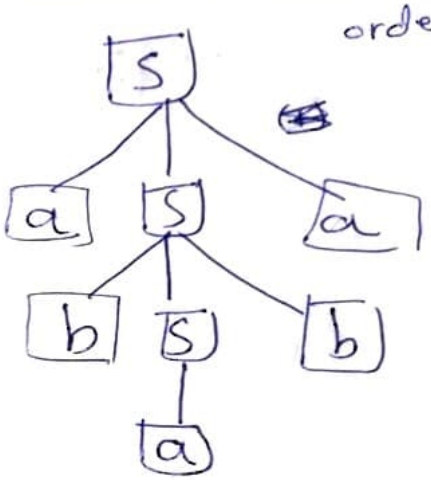
Derivation of aab →

$aa\lambda\lambda b = aab$

Example (2)

G_3 1. $S \to aSa$
2. $S \to bSb$
3. $S \to a$
4. $S \to b$
5. $S \to \lambda$

order: 1,2,3



Derivation of ababa

$S \overset{(1)}{\Rightarrow} aSa$
$S \overset{(2)}{\Rightarrow} abSba$
$S \overset{(3)}{\Rightarrow} ababa$

$$L(G) = \{w : w \in T^*, S \overset{*}{\Rightarrow} w\}$$

Example:

G:

(1) or (2) (3)

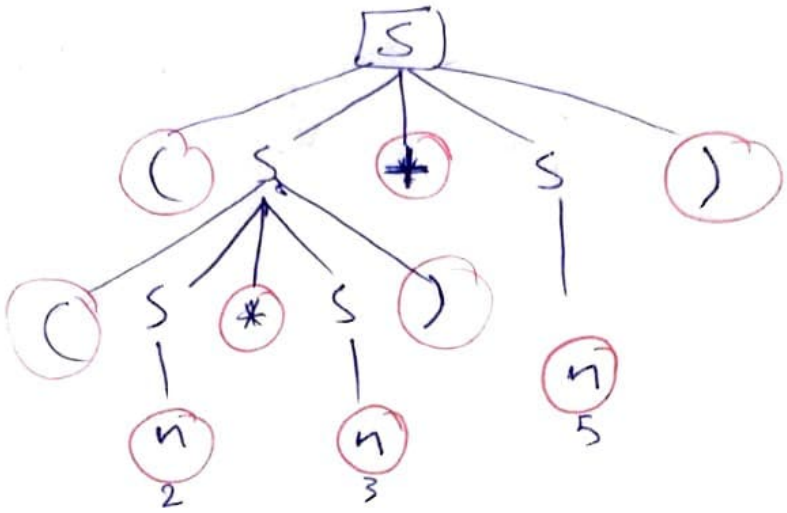
$S \rightarrow (S+S) \mid (S*S) \mid n$

$S \rightarrow a \}$
$S \rightarrow b \}$ $\rightarrow a|b$

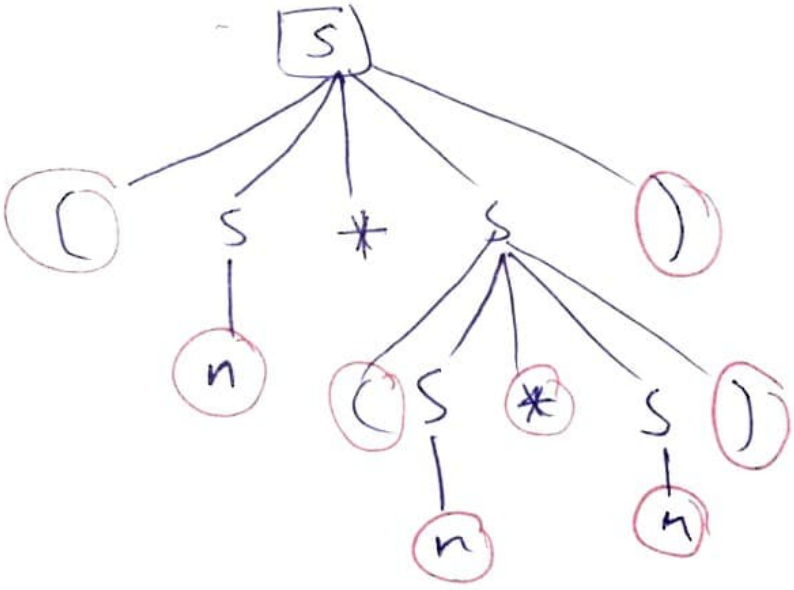
يعني 2 rule

* Derivation of ((2*3)+5)

((n*n)+5



* Derivation of (n*(n*n)



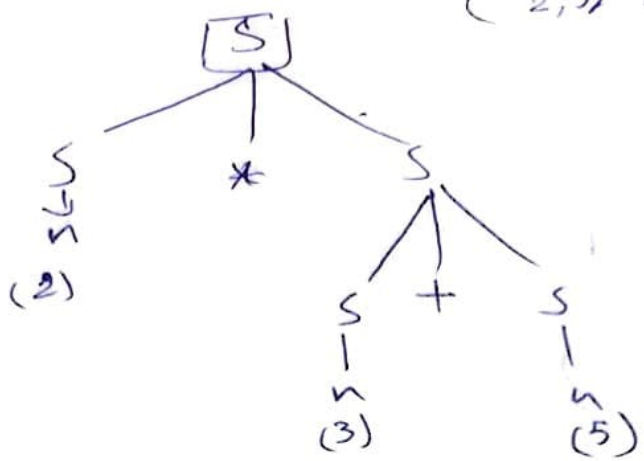
Example (1) (2) (3) (Ambiguous)

G: $S \rightarrow S+S \mid S*S \mid n$

= 11

* Derivation of String "2*3+5" (parse tree)

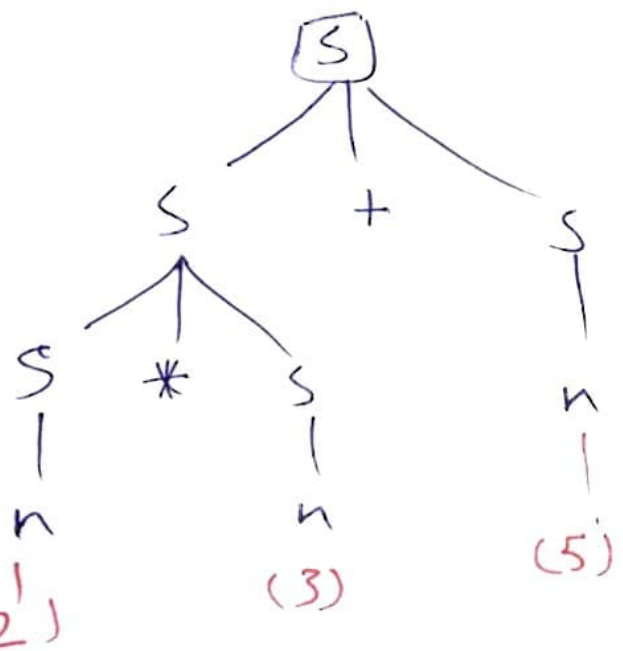
(1) (2,3,1,3,3) order of rules



الجواب = 16
غلط

هون حجمنا بعيد عن حزبنا
وصلنا بالأول لولبات غلط

(2) (1,2,3,3,3) order of rules



هون بضرب بعدين
بجمع
الجواب 11
صح

* Ambiguous ⇒ 2 derivation لـ string نفس ↓ Different

يعني نفس ال string لها Derivations مختلفة

Examples: Ambiguous Grammar.

G: stmt → if stmt | for stmt | while stmt

If_stmt → if Expr then stmt | if Exp then stmt else stmt

Expr → Conditional-exp | Arithmetic exp

```
if cond then
    if cond then
        stmt
else
        stmt
```

---

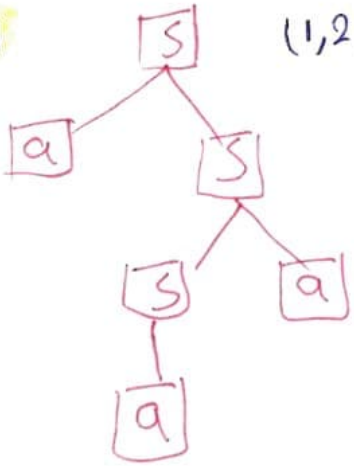
* Ambiguity

Derivation لكل طريق كتير لـ rule - انه نقدر لكل انه نستبدل -

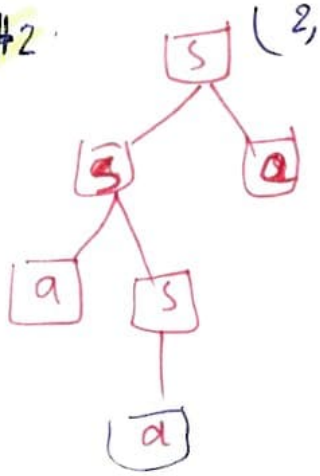
ex: G: 1. S → aS
2. S → Sa
3. S → a

① Derivation of aaa

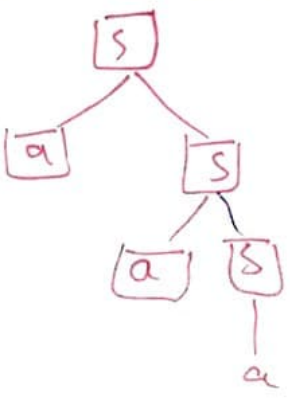
#1 (1,2,3) order



#2 (2,1,3)



#3 (1,1,3) order



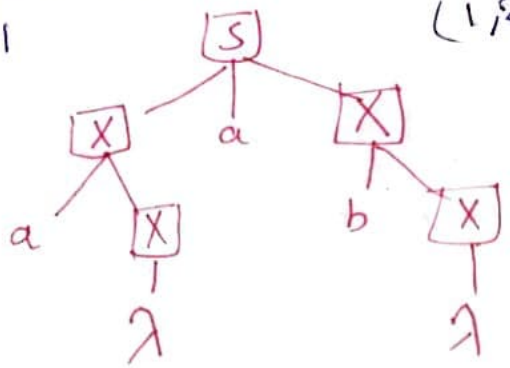
Example:

G: 1. $S \rightarrow XaX$

2. $X \rightarrow aX \mid bX \mid \lambda$ ③ ④

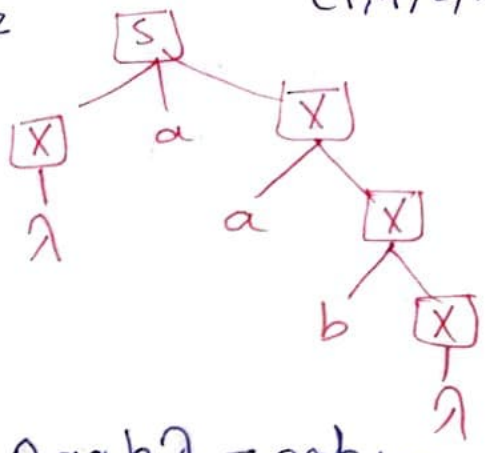
① Derivation of aab

#1 (1,2,4,3,4)



$a\lambda ab\lambda = aab$ ✓

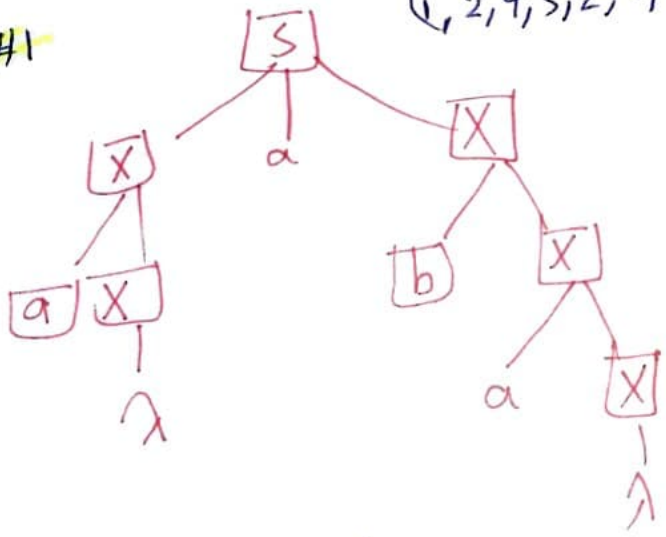
#2 (1,4,2,3,4)



$\lambda aab\lambda = aab$ ✓

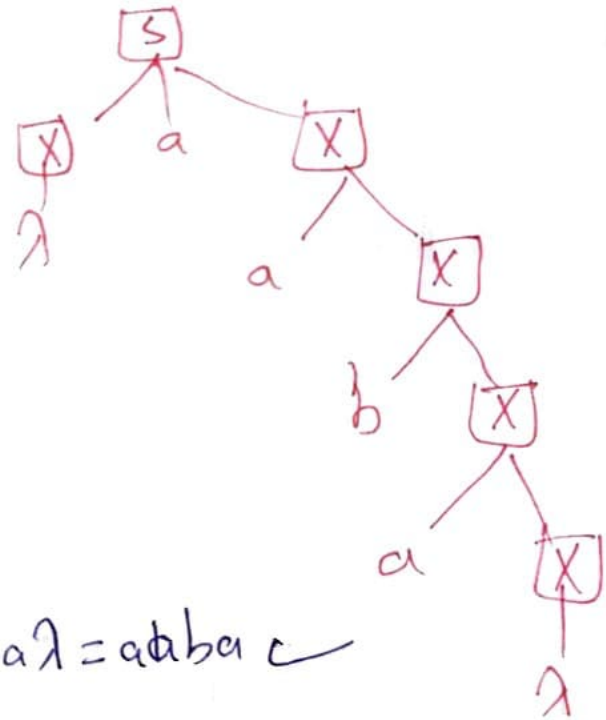
② Derivation aaba

#1 (1,2,4,3,2,4)



$a\lambda aba\lambda = aaba$ ✓

#2



$\lambda aaba\lambda = aaba$ ✓

Example

$G: S \to \overset{(1)}{aSb} \mid \overset{(2)}{bX}$ $\quad X \to \overset{(3)}{bX} \mid \overset{(4)}{b}$

Does this Grammar Ambiguous?

① Derivation of abbb

$S \overset{(1)}{\Rightarrow} aSb$

$S \overset{(2)}{\Rightarrow} abXb$

$S \overset{(4)}{\Rightarrow} abbb$ ✓

② Derivation of bb

$S \overset{(2)}{\Rightarrow} bX$

$S \overset{(4)}{\Rightarrow} bb$ ✓

$L(G) = \{ bb, bbb, bbbb, abbb, abbbb, aabbb, \dots \}$

$L(G) = \{ a^n b^k b^n : k \geq 2, n \geq 0 \}$

---

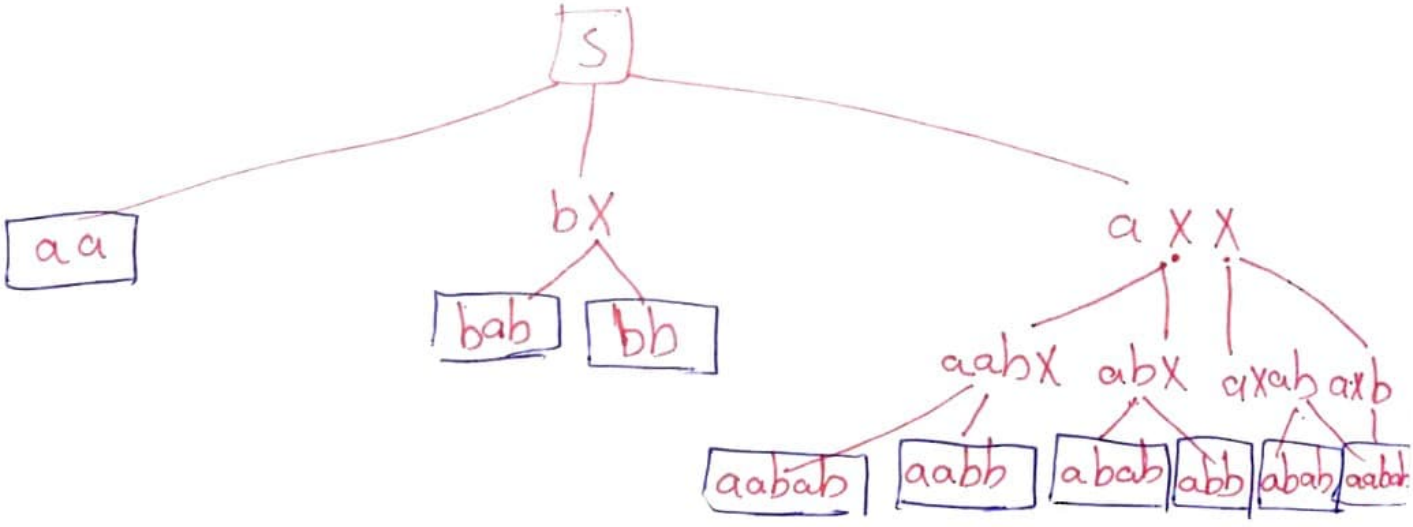
* Total Language Tree:

كل strings ال Language عبر عن

example:

$S \to aa \mid bX \mid aXX$

$X \to ab \mid b$

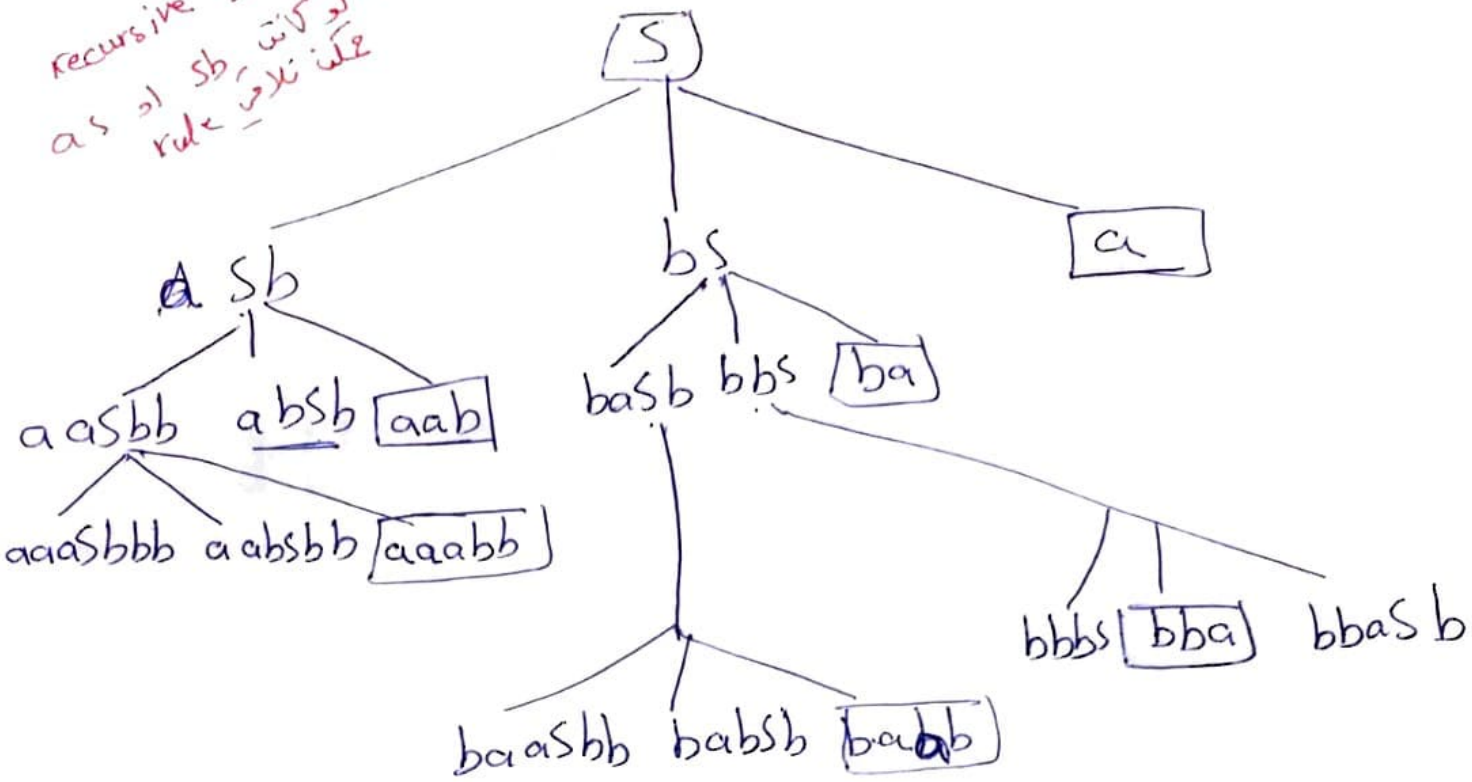


Language = {aa, bab, bb, aabab, aabb, abab, abb, abab, aabab, aabb, abb}

Example 3

$S \rightarrow aSb \mid bS \mid a$

recursive — لو كانت a او b على تلاقي rule



وهكذا نبقى نفرع ، وهيب الاخر rule حتى

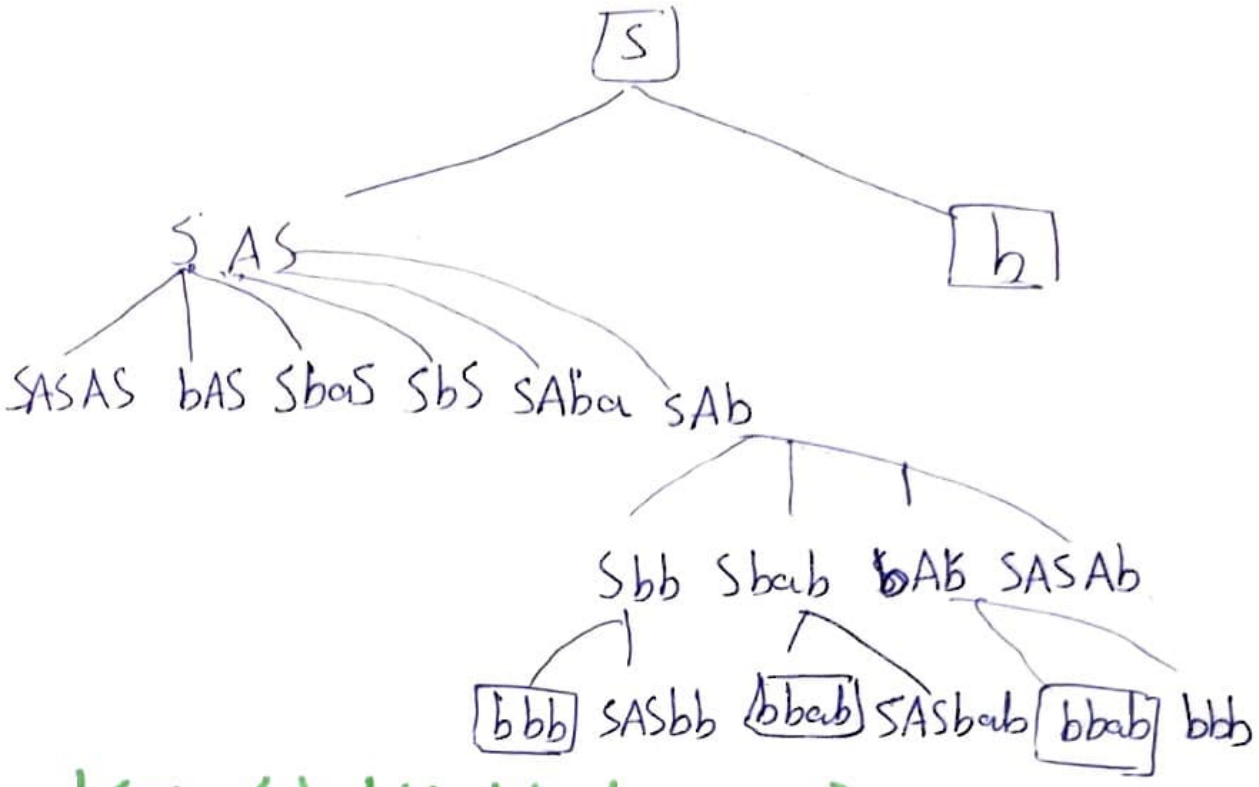
$L(G) = \{a, ba, aab, bba, baab, abab, aaabb, \ldots\}$

. . . context free language هاي اسمها

Example:

G: $S \to SAS \mid b$ (2 recursive call)

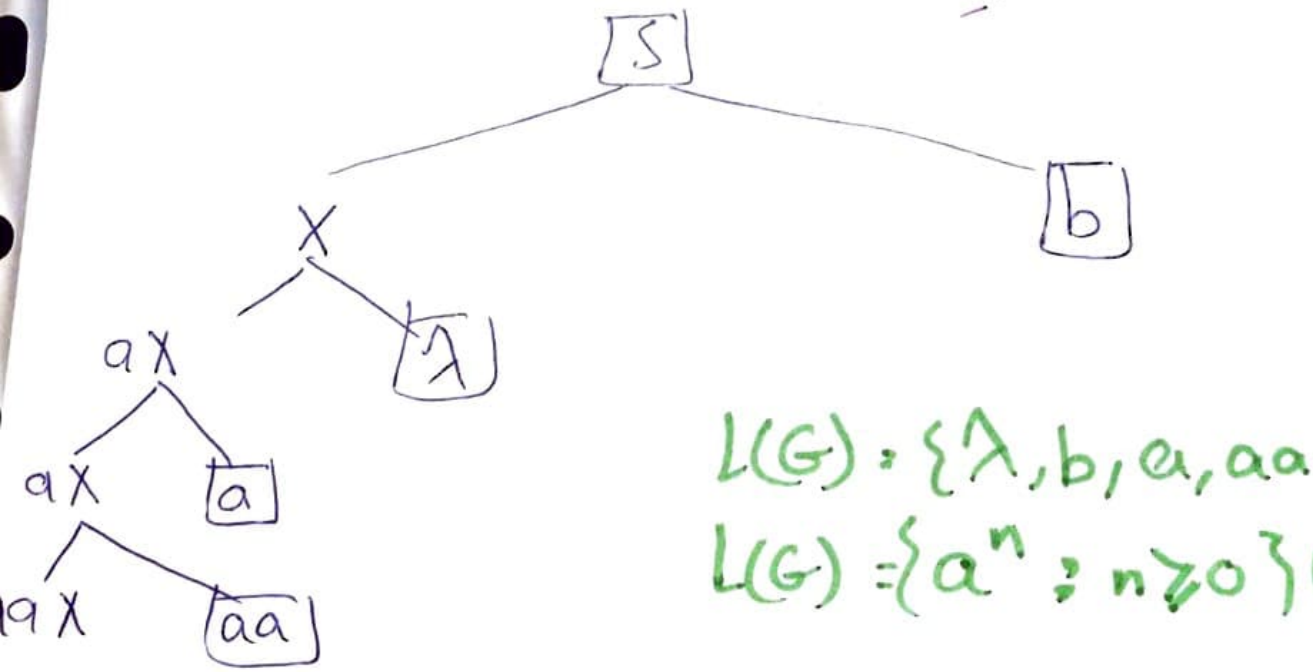
$A \to ba \mid b$



$L(G) = \{b, bbb, bbab, \dots\}$

Example 3:

G: $S \to X \mid b$ $\quad X \to aX \mid \lambda$ (إذا شفنا λ نطلع عالقاعدة اللي قبلها rule)

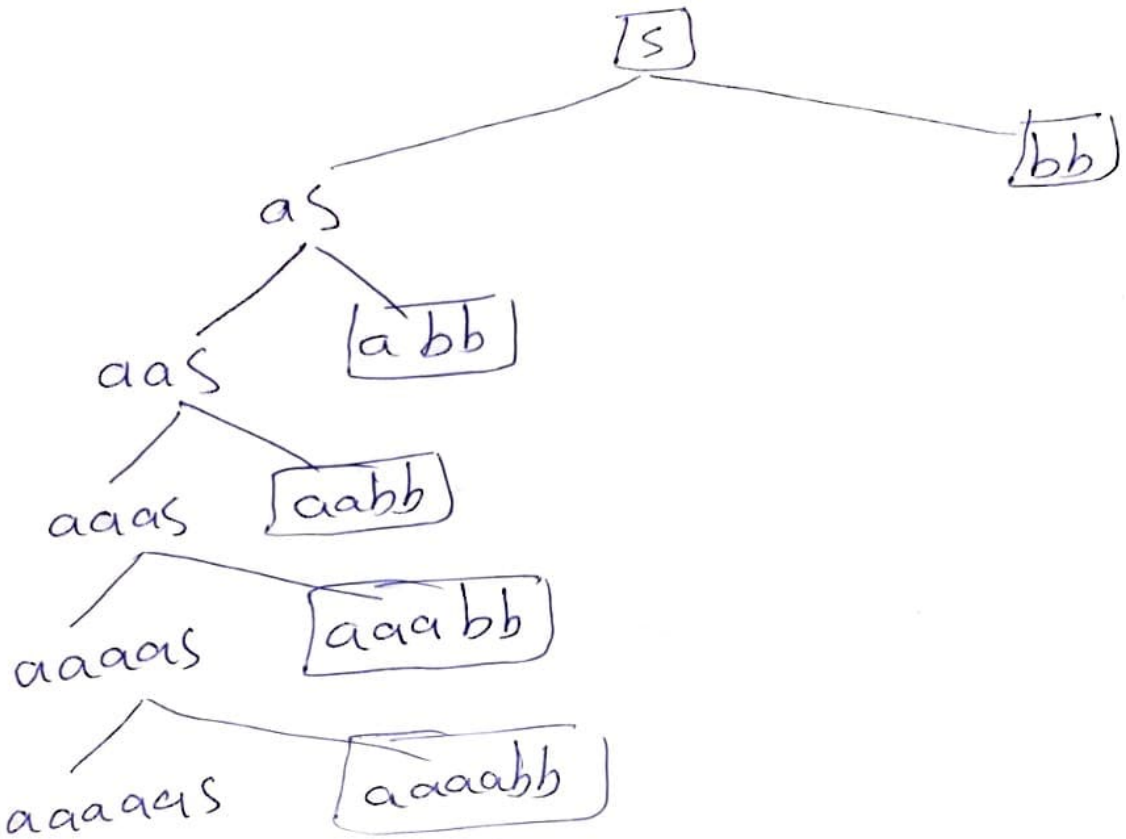


$L(G) = \{\lambda, b, a, aa, aaa, \dots\}$

$L(G) = \{a^n : n \geq 0\} \cup \{b\}$

Example: Define final L(G) using total languages tree

G: $S \to aS$
$S \to bb$



$L(G) = \{bb, abb, aabb, aaabb, aaaabb, \ldots\}$

$L(G) = \{a^n bb : n \geq 0\}$

$r = a^* bb$

Example 1

assume CFG:

G: $S \to XYX$
$X \to aX \mid bX \mid \lambda$
$Y \to bbb$

X Y X
aX, bX, λ bbb aX, bX, λ
$\{a,b\}^*$ bbb $\{a,b\}^*$

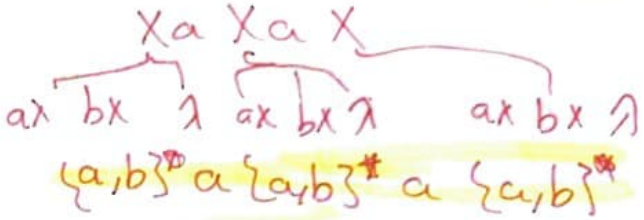
$L(G) = \{bbb, abbb, abbbb, bbba, bbbb, \dots\}$

$L(G) = \{w_1 bbb w_2 : w_1, w_2 \in \{a,b\}^*\}$

RegEx: $(a+b)^* bbb (a+b)^*$

Example 2

$S \to XaXaX$
$X \to aX \mid bX \mid \lambda$



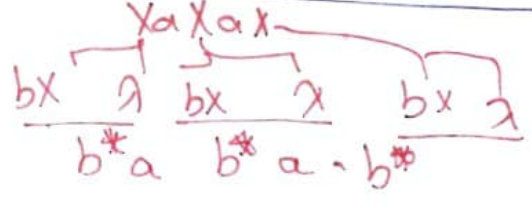
$L(G) = \{aa, aaa, baa, aab, aba, \dots\}$

$L(G) = \{w_1 a w_2 a w_3 : w_1, w_2, w_3 \in \{a,b\}^*\}$

RegEx: $(a+b)^* a (a+b)^* a (a+b)^*$

Example 3

$S \to XaXaX$
$X \to bX \mid \lambda$



$L(G) = \{aa, baa, aab, aba, aabb, \dots\}$

$L(G) = \{b^n a b^m a b^k : n, m, k \geq 0\}$

RegEx: $b^* a b^* a b^*$

Example 4

$G: S \to XaXaX \mid SS$

$X \to bX \mid \lambda$

$L(G) = \{(b^n a b^m a b^k)^r ; n,m,k \geq 0, r \geq 1\}$

$RegEx = (b^* a b^* a b^*)^+$

Example 5

$G: S \to SS \mid XaXaX \mid \lambda$

$X \to bX \mid \lambda$

$L(G) = \{bb, aab, \ldots\}$

$L(G) = \{(b^n a b^m a b^k)^r : n,m,k,r \geq 0\}$

$RegEx = (b^* a b^* a b^*)^*$

Example 6

$S \to abb \mid aXbb$

$X \to aX \mid bX \mid \lambda$

$L(G) = \{abb, aabb, \ldots aaabbbbb, \ldots\}$

~~$L(G) = \{a^n a^m b^m \ldots\}$~~

$L(G) = \{a^n w b^{2n} : w \in \{a,b\}^* ; n \geq 1\}$

Example 7:

$S \to aSb \mid aXb$

$X \to aX \mid \lambda$

$L(G) = \{ a^n a^m b^n , n \geq 1, m \geq 0 \}$

Example 8:

$S \to XYZX$

$X \to aXb \mid \lambda$

$Y \to yaa \mid \lambda$

$Z \to aZ \mid bZ \mid \lambda$

$X : a^n b^n$

$y : (a)^{2k}$ $k \geq 0$

$Z : w : w \in \{a,b\}^*$

$L(G) = \{ a^n b^n a^{2k} w a^m b^m , n,m,k \geq 0, w \in \{a,b\}^* \}$

#Regular Grammar.

Regular Grammar Generates Regular Language

Right Linear Grammar

$A \to xB$ (الرمز على اليمين)

$A \to x$

Left linear Grammar.

$A \to Bx$ (الرمز على اليسار)

$A \to x$

Example 1

$S \rightarrow abS \mid a$ Right linear Grammar.

$L(G) = \{a, aba, ababa, ababababa, \dots\}$

Pattern $L(G) = \{(ab)^n a \;,\; n \geq 0\}$

RegEx $= r = (ab)^* a$

Example 2

Right and Left linear Grammar

$S \rightarrow A$
$A \rightarrow aB \mid \lambda$
$B \rightarrow Ab$

} transform
$S \rightarrow A$
$A \rightarrow aAb \mid \lambda$
} CFG

$L(G) = \{\lambda, ab, aabb, aaabbb, aaaabbbb, \dots\}$

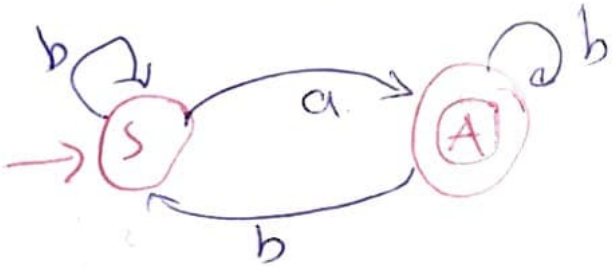
RegEx يعبر عن لغة Reg Grammar إما Left أو Right

$L(G) = \{a^n b^n \;,\; n \geq 0\}$

*Convert FA to Regular Grammar:-



	a	b
-So	S1	So
+S1	So	S1



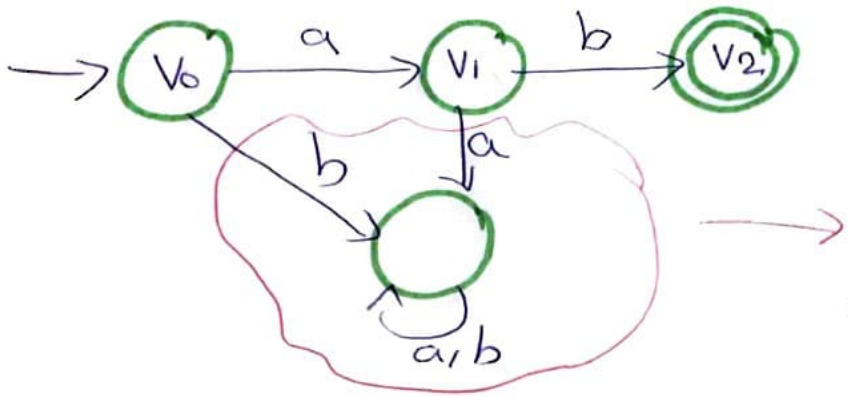
$S \rightarrow bS \mid aA$
$A \rightarrow bA \mid aS \mid \lambda$

- أول خطوة نعين متغير لكل variable
- كل rule نكتبها linear
- كل transition rule, نعمل rule جديدة goes to λ وهي Final State

In General

- $\delta(q_0, u) \rightarrow q_1 \Rightarrow q_0 \rightarrow uq_1$

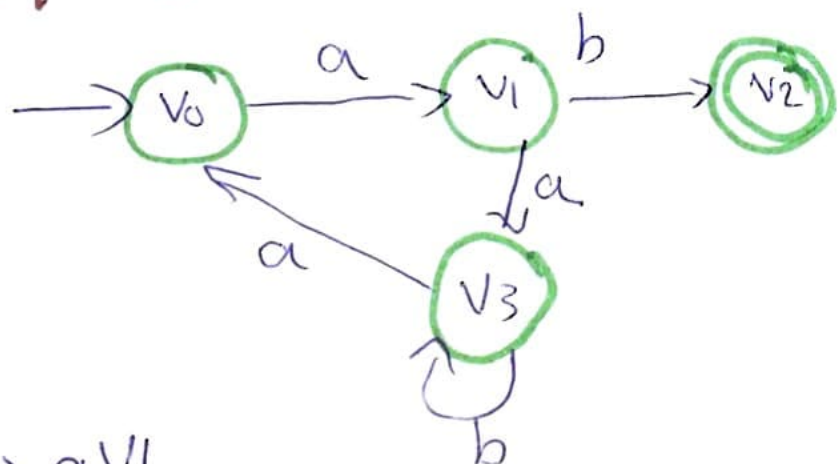
Example 1



هذا يُرد
ما يتطابق عليه
rejected

$V_0 \rightarrow aV_1$
$V_1 \rightarrow bV_2$
$V_2 \rightarrow \lambda$

Example 2



accept يقبل

$V_0 \rightarrow aV_1$
$V_1 \rightarrow bV_2$
$V_1 \rightarrow aV_3$
$V_3 \rightarrow aV_0$
$V_3 \rightarrow bV_3$
$V_2 \rightarrow \lambda$

Example:

Write a Regular Grammar for the following Languages:

① $L = \{a^n b : n \geq 1\}$

A

$S \rightarrow Ab$

$A \rightarrow aA \mid a$

② $L = \{a^n b^m c^k : n \geq 1, m \geq 2, k \geq 0\}$

$A \quad B \quad C$

$S \rightarrow ABC$

$A \rightarrow aA \mid a$

$B \rightarrow bB \mid bb$

$C \rightarrow cC \mid \lambda$

③ $L = \{ba^{2n} : n \geq 1\}$

$A \quad B$

$S \rightarrow AB$

$A \rightarrow b$

$B \rightarrow baaB \mid aa$

* Simplifications of context-free Grammars :-

① Substitution Rule : $(A \to xBz \xrightarrow{\text{Substitute } B \to y} A \to xBz \mid xyz$

Example :

الهدف اني الغي أحد (Variable)

الـ A ما بقدر الغيه لانه بفرلستي تفسه

$S \to aB$
$A \to aaA$
$A \to abBc$
$B \to aA$
$B \to b$ — Substitute $B \to b$

يعني بعوض مكان كل $B \Leftarrow b$

$S \to aB \mid ab$
$A \to aaA$
$A \to abBC \mid abbc$
$B \to aA$ — Substitute $B \to aA$

* صرنا بعوض مكان كل $B \Leftarrow aA$

آخر خطوة بحذف الـ B اشي فيه

$S \to aB \mid ab \mid aaA$
$A \to aaA$
$A \to abBc \mid abbc \mid abaAc$

Final answer :

$S \to ab \mid aaA$
$A \to aaA \mid abbc \mid abaAc$

② Nullable Variables

λ-Production

$X \to \lambda$

Nullable variable

$S \to A$
$A \to \lambda$

Example-

$S \to aMb$
$M \to aMb$
$M \to \lambda$

remove λ-Production ⟹

$S \to aMb \mid ab$
$M \to aMb \mid ab$

λ-Production — Nullable Variable

③ Unit production:

$X \to Y$

Variables.

Example:

$S \to aA$
$A \to a$
$A \to B$ ← unit production
$B \to bb$

substitute $A \to B$ ⟹

$S \to aA \mid aB$
$A \to a$
$B \to A \mid B$ ~~B~~
$B \to bb$

substitute $B \to A$ ⟹

$S \to aA \mid aB \mid$ ~~aA~~
$A \to a$
$B \to bb$

Final rule

remove unit production of a form $X \to X$

$B \to B$ ✗

لأنه ما بيفيد بشي

(4) Useless Production.

Example

$S \to aSb$

$S \to \lambda$

$S \to A$
$A \to aA$

useless Production
طرف لا يتوقف ولا ينتهي

$S \Rightarrow A \Rightarrow aA \Rightarrow aaA \Rightarrow aaaA$

$B \to \alpha S$

unreachable
لا يمكن الوصول إليه من Start

---

*Removing Useless Variables and Production

Example:

$S \to aSb$

$S \to \lambda$

$S \to A$
$A \to aA$
$B \to C$
$C \to D$

→ useless production

useless Variables

useful

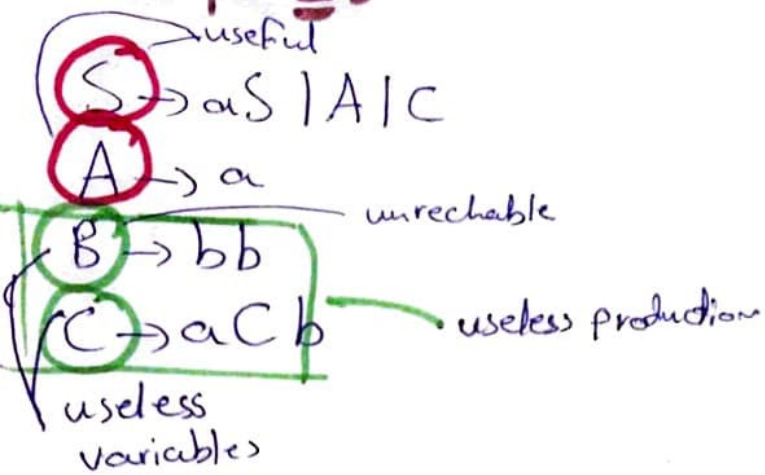
$S \Rightarrow \cdots \Rightarrow xAy \Rightarrow \cdots \Rightarrow w$

usefull

$w \in L(G)$

String

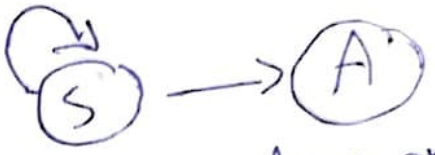
Example 2:



Final rules

$S \to aS \mid A$

$A \to a$



Dependency graph

Example 3:

$$\left.\begin{array}{l} S \to \lambda \\ S \to aaSbb \end{array}\right\} \Rightarrow L(G) = \{\lambda, aabb, aaaabbbb\}$$

remove λ

$$\left.\begin{array}{l} S \to aaSbb \mid aabb \end{array}\right\} \Rightarrow L(H) = L(G) - \{\lambda\}$$

① chomsky Normal form (CNF):

Variable

$A \to BC$

$A \to a$

terminal

* همیشه شکل باید لازم یکون

2 variable
1 terminal
یا
1 terminal

* هر جا فی rule بترجح از ۲ تا کلها strings

Example:

$S \to aSb$

$S \to b$

A B

Normal شی
خشان لازم
یکون فیا
2 Variables
chomsky لازم

$S \to ASB$

$S \to B$

$A \to a$

$B \to b$

X

$\Rightarrow$

$S \to AX$

$X \to SB$

$A \to a$

$B \to b$

$S \to b$

Example:

$S \to ABa$

$A \to aab$

$B \to Ac$

$\Rightarrow$

$S \to ABTa$ V1

$A \to TaTaTb$ V2

$B \to ATc$

$Ta \to a$

$Tb \to b$

$Tc \to c$

$\Rightarrow$

$S \to AV1$

$V1 \to BTa$

$A \to TaV2$

$B \to ATc$

$Ta \to a$

$Tb \to b$

$Tc \to c$

$V2 \to TaTb$

هر terminal one variable

Chomsky: CNF

$A \to C_1 C_2 \ldots C_n$

$$\Downarrow$$

$A \to C_1 V_1$
$V_1 \to C_2 V_2$
$V_2 \to C_3 V_3$
$V_3 \to C_4 V_4$
$V_{n-2} \to C_{n-1} C_n$

(2) Greinbach Normal Form (GNF)

Example: 1

$S \to cAB$
$A \to aA \mid bB \mid b$
$B \to b$

Example: 2

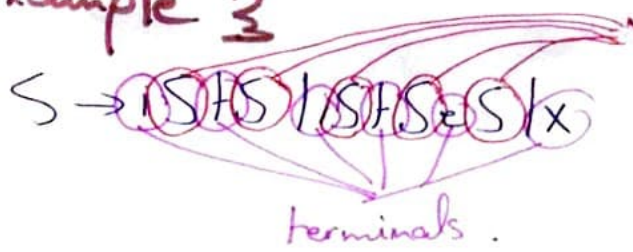
$S \to abSb$
$S \to aa$

$\Rightarrow$

$S \to aT_bST_b$
$S \to aT_a$
$T_b \to b$
$T_a \to a$

Normal Greinbach.

Example 3



if-else statement

$S \to iST_fS$

$S \to iST_fST_eS$

$S \to x$

$T_f \to t$

$T_e \to e$

---

* write a Grammar for the following language:

① $L = \{a\}$

G: $S \to a$

② $L = \{a^n b^m : n \geq 1, m \geq 2\}$

X Y

G: $S \to XY$

$X \to aX \mid a$

$Y \to bY \mid bb$

③ $L = \{a^n : n \geq 1\}$

G: $S \to aS \mid a$

④ $L=\{c\underbrace{a^k}_{X}\underbrace{b^j}_{Y}a \mid k\geq 0, j\geq 3\}$

~~$S\to cXYa$~~

$G: S\to cXYa$

$X\to aX \mid \lambda$

$Y\to bY \mid bbb$

⑤ $L=\{a^{2n} \mid n\geq 0\}$

$G: S\to aaS \mid \lambda$

⑥ $L=\{b^{3n} \mid n\geq 1\}$

$G: S\to bbbS \mid bbb$

⑦ $L=\{b^{2n}a^{3m}c \mid n\geq 1, m\geq 0\}$

$G: S\to AB\mathbf{C}$ — Csmall

$A\to bbA \mid bb$

$B\to aaaB \mid \lambda$

⑧ $L=\{\underbrace{a^n b^n}c \mid n\geq 0\}$

لا يكون نفس الحرف يتبع فقط one variable

$G: S\to Xc$

$X\to aXb \mid \lambda$

(9) $L = \{a^n b^n : n \geq 1\}$

$G: S \to aSb \mid ab$

(10) $L = \{a^n c b^n : n \geq 1\}$

$G: S \to aSb \mid acb$

(11) $L = \{a^n c b^n : n \geq 0\}$

$G: S \to aSb \mid c$

(12) $L = \{a^n x b^n c : n \geq 0\}$

$G: S \to XC$

$X \to aXb \mid X$